A Comparison of Spatial and Temporal Models Conducted on County Level Drug Overdose Rates and the Interaction of Various Socioeconomic Variables

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Abstract

Drug overdoses are a growing concern in the USA. I investigated the spatial and temporal patterns of drug overdoses, as well as the possible interaction of socioeconomic variables. I used the restricted files from the National Center of Health Statistics (NCHS) to get county level overdose rates per 100K people. I ran many models to capture these relationships, to include random walk with drift, spatial autoregression (SAR), random forest for space and for time, and Gaussian Process Boosting (GPBoost). Of the socioeconomic variables studied, median household income and poverty estimates were the most important variables for these models.

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1 Introduction

Drug overdoses are a growing concern is the US. From 2005 to 2023 over a million people died from drug overdoses compared to 703,684 who died by firearms and 734,840 who died by vehicles (Centers for Disease Control and Prevention, 2026b). Deaths caused by drug overdoses have seen a steep increase over the last 20 years. Speculation about the potential causes of this growing crisis are numerous, but without any clear direction to fixing this issue. Socioeconomic factors may play a roll in determining what counties are suffering from high overdose rates. What remains curious is these socioeconomic factors are not correlated the same with the overdose rate by all regions. For this project I will investigate the spatial and temporal relationships of drug overdoses as well as potential causes across the contiguous United States and Washington DC at the county level from 2005 to 2023.

1.1 Background Studies

In a paper entitled “Identifying Counties at Risk of High Overdose Mortality Burden During the Emerging Fentanyl Epidemic in the USA: a Predictive Statistical Modeling Study”, the authors studied county level overdose data from 2010 to 2018 for the contiguous USA (Marks et al., 2021). The main goal of this study was to identify counties at risk for high overdose death rates. The authors did this by running a negative binomial regression model with random intercepts by county and random slopes by year. Each model was trained starting in 2010 up to a year before the test year, with a minimum of two years to train on. So, the first model has a training set of 2010 to 2011, and a test set of 2012, then predicts 2013. Each prediction for 2013 to 2018 is from a different negative binomial model following the setup above to make the training and test sets. Additionally, the model included various predictor variables as fixed effects. The models in this article outperformed a benchmark model. The benchmark model predicted using the slope of change in overdose death rates in the same years as the training set. This paper used the restricted county level data from the National Center of Health Statistics.

In the article "Accurate Spatiotemporal Mapping of Drug Overdose Deaths by Machine Learning of Drug-Related Web-Searches" the authors studied the relationship between drug related searches and drug overdoses (Campo et al., 2020). They used random forest to predict yearly estimates at the state and county levels of overdose deaths as well as monthly estimates at the state level using data from 2004 to 2017. This paper used NCHS data attained through the Center For Disease Control Wide-ranging ONline Data for Epidemiologic Research (CDC WONDER), which is publicly available data and therefore highly suppressed at the county level. They tested many machine learning methods, including random forest. They found high correlation between search terms associated with drug overdoses and overdose deaths rate, which is dependent on location.

In the paper "A Spatial Gaussian-Process Boosting Analysis of Socioeconomic Disparities in Wait-Listing of End-Stage Kidney Disease Patients across the United States" the authors used Gaussian process boosting (GPBoost) to understand the spatial patterns of wait-listing for kidney transplants and the potential interaction with socioeconomic data (Chakraborty et al., 2024). GPBoost was used to incorporate a Gaussian process for the spatial effects and fixed effects for covariate data.

1.2 Overview of Models

To further investigate the overdose deaths from a spatial and temporal perspective I will focus on a number of models. Some of these models specialize in spatial and/or temporal data, with some having the ability to incorporate fixed and random effects from covariates. The models I chose to use are: random walk with drift, spatial autoregression (SAR), random forest, and Gaussian process boosting (GPBoost).

Random walk with drift is a stochastic process where the next step is the previous step plus some random error, and drift adds a constant term to each step. So this drift pushes the random error in some preferred direction. This process is considered a very basic way to model time series data. I use this model to serve as a benchmark that all other models are compared to.

Spatial autoregression (SAR) is commonly used to analyze data with spatial correlation. It does this by taking into account that observations that are close together are more correlated than observations that are further apart. This is an autoregression model designed to handle spatial data instead of temporal data. Instead of handling distance in time, it handles distance in 2-space. This type of model is good for analyzing data with inherit spatial relationships.

Random forest is a machine learning technique that uses decision trees to make predictions. Random forest is commonly used for regression and classification tasks. Random forest can also be used for time series and spatial data. To use random forest for temporal data and forecasting I followed code from Tilgner (Tilgner, 2019). This method only accounts for temporal effects. To use random forest for spatial data I followed code from Nowosad (Nowosad, 2025). This method only accounts for spatial effects.

To incorporate the effects of space and time, I decided to use Gaussian process boosting (GPBoost). GPBoost can model the space or time effect with a Gaussian process while incorporating fixed and random effects. A complete review of the capabilities of GPBoost was written by Sigrist in a published paper called "Gaussian Process Boosting" (Sigrist, 2022). I also followed some code demonstrations from Sigrist. To include a Gaussian process for the spatial effect I referenced "Mixed Effects Machine Learning with GPBoost for Grouped and Areal Spatial Econometric Data" by Sigrist (Sigrist, 2023b). To include a Gaussian process for the temporal effect I referenced "Mixed Effects Machine Learning for Longitudinal & Panel Data with GPBoost Part III" by Sigrist (Sigrist, 2023a). Together these descriptions allowed me to build out the models I present in section 3.

In section 3 I will fit each of the models discussed here only using data in the state of West Virginia, and I will compare the results of these models. Then I will divide the contiguous states into 8 regions and fit two types of GPBoost models to each of these 8 regions in the years 2005 to 2022, and then compare them.

2 Overdoses by County 2005 to 2023

2.1 Mortality Data from the National Center for Health Statistics

The county level overdose data comes from the Detailed Mortality All Counties files from the Division of Vital Statistics, National Center for Health Statistics (NCHS). Each observation is an individual's mortality record, across the years 2005 to 2023. Since these records are individual patient records, the public use files are aggregated over an area and suppressed if the counts are between 1 and 9 or if the population is below 10. The public use files can be attained from Center For Disease Control Wide-ranging ONline Data for Epidemiologic Research (CDC WONDER) (Centers for Disease Control and Prevention, 2026a). The public use data at the state level is not missing much data, since in this case it is aggregate by state. The state level overdose rate does not show a detailed enough spatial pattern but the county level rate does. The county level public use data is highly suppressed, due to low death counts in a county or due to small population. So, for this analysis, I acquired the restricted county level data files from NCHS by going through the application process detailed on their website (National Center for Health Statistics, 2022).

In the data, Federal Information Processing Standards (FIPS) codes are used to identify the county and state where each person died. The 10th revision of the International Statistical Classification of Diseases, Injuries, and Causes of Death (ICD-10) code is used to identify what caused each individual's death. There are also various demographic characteristics included about each person. A FIPS code is a 5-digit code, with the first two digits identifying the state, and the last three identifying the county. There are over 3,000 counties in the USA. The key is publicly available here (Centers for Disease Control and Prevention, 2024a). ICD-10 is used to identify the cause of death

(Centers for Disease Control and Prevention, 2018; National Center for Health Statistics, 2025). For this analysis I used ICD-10 codes X40 to X49 and X60 to X69.

In addition to the location and date of death the following information is also included:

- Education (9) levels: 8th grade or less, 9-12th grade but no diploma, high school graduate or GED completed, some college credit but no degree, Associate degree, Bachelor's degree, Master's degree, Doctorate or professional degree, Unknown.
- Sex (2) levels: Male, Female.
- Marital status (7) levels: Never married, Married, Widowed, Divorced, Marital Status not on certificate, Marital Status unknown.
- Age (6) levels: 0-14, 15-24, 25-44, 45-64, over 64, not listed. I defined these levels from the exact age given in the data.
- Race (14) levels: White, Black, American Indian (includes Aleuts and Eskimos), Asian, etc... The issue here is the race codes kept being redefined, so no single code spans all data years 2005-2024, and none are compatible.
- Population (5) levels: County of 1,000,000 or more, County of 500,000 to 1,000,000, County of 250,000 to 500,000, County of 100,000 to 250,000, County of less than 100,000. I gather more accurate population data from ACS.

2.2 Population and Covariate Data from the American Community Survey

The population data supplied by the main data set are in ranges, so I instead used the estimates provided by the Census Bureau through the American Community Survey (ACS), in Table B01003. ACS 1-year estimates do not include a county's population if it is below 65,000 people whereas ACS 5-year estimates are not restricted (Centers for Disease

Control and Prevention, 2020). However, ACS did not complete 5-year estimates for the population data in years 2005 to 2009. Meaning that 2005 to 2009 has a lot of missing population data at the county level. I used ACS 1-year estimates for 2005 to 2009, and ACS 5-year estimates for 2010 to 2023.

I also collected some covariate data from ACS Tables S1810, S2701, S1501, S1201 and S1901. Table S1810 has estimates of various disabilities in a county. I focused on cognitive disabilities (Cognitive) and self-care (SelfCare) disabilities. Table S2701 has estimates of people with health insurance (Insured). Table S1701 has estimates of poverty status (Poverty). Table S1501 has estimates of the highest level of education attained. I focused on high school education and below (Education). Table S1201 has estimated percents of marital status by county; such as married, divorced, etc. I focused on the divorced rate (maritalStatus_D_ACS). Table S1901 has estimates of median household income (medianIncome). All of these tables are 5-year estimates from 2015 to 2023. I collected ACS data using the tidycensus package in R (Walker & Herman, 2025).

2.3 Data Cleaning

If a county was not listed in a given year, I added that county back into the data set with a zero death count. There were a few counties that underwent a name change, which I had to account for when running models across space and time. In South Dakota, Shannon County (FIPS: 46113) was renamed, Oglala Lakota County (FIPS: 46102), though its boundaries remained unchanged (Centers for Disease Control and Prevention, 2024b). I resolved this by treating the codes as the same, since they practically are. In 2022 Connecticut changed from having 8 counties to a county equivalent system with 9 planning regions, so the FIPS codes changed, and the boundaries changed (U.S. Census Bureau, 2022). This is a problem because ACS population data was updated to reflect this change, but NCHS mortality data was not. Meaning, NCHS still reports the location of death in the legacy county boundaries, but ACS reports its data in the new planning regions. This makes it difficult to run analysis on Connecticut across all years, 2005 to 2023. So, Connecticut is not included in this analysis for the years 2022 onward. Bedford City, Virginia (FIPS: 51515) was an independent city until 2015 when it joined Bedford County (FIPS: 51019) in 2013 as a town (Centers for Disease Control and Prevention, 2024b). I added 51515's data to 51019 for all years.

Missing overdose rate data is often the result of ACS suppressing the population data for a given county in that year. For Connecticut the missing data occurs in the years 2022 to 2023 because NCHS mortality data and ACS data at the county equivalent level are recorded within different boundary lines. In final reports, any rates where the counts are between 1 and 9 are suppressed, per NCHS data use agreement.

2.4 Exploratory Analysis

The overdose rate, regardless of region, is generally increasing at the same rate year to year. West Virginia (when ignoring DC) has had the highest drug overdose rate since 2010. Starting at 27.71 deaths per 100K people in 2010, peaking at 84.4 deaths per 100k people in 2021, and then decreasing to 79.02 deaths per 100K in 2023, Figures 1 and 2. What is also interesting is that the state second to WV is not a neighboring state, it is usually NM or DC. The trick with comparing DC here is that it is more on the level of a city or a county and not of a state. But NM being second to WV in many years has no clear spatial reasoning.

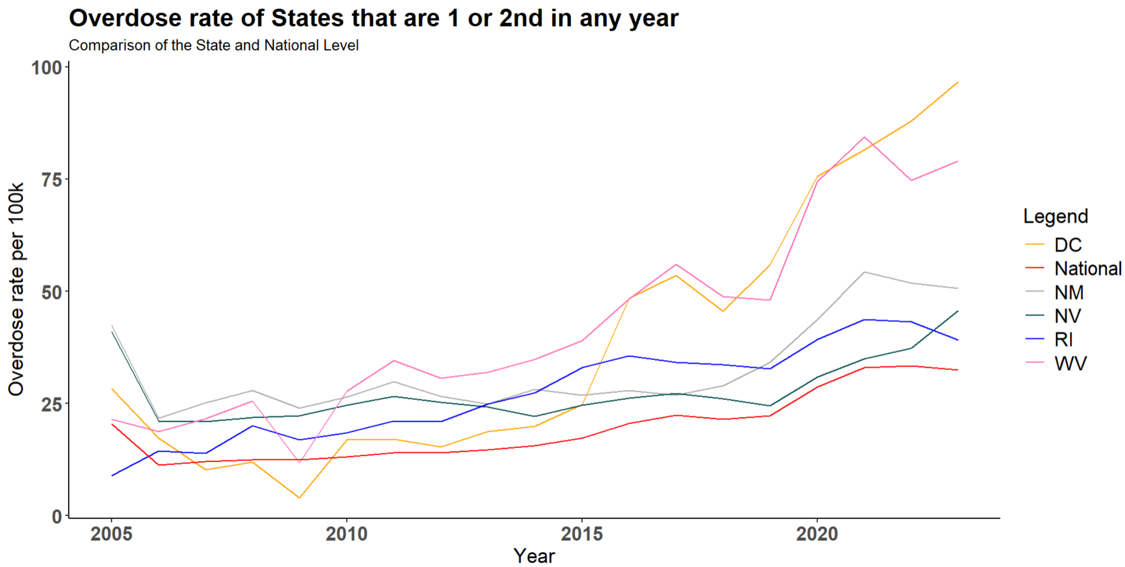


Figure 1: Line plot of states with the highest or second-highest overdose rates in each year.

Highest			Second Highest		
Year	**State**	**Rate**	**Year**	**State**	**Rate**
2005	NM	42.39	2005	NV	40.99
2006	NM	21.69	2006	NV	21.16
2007	NM	25.13	2007	WV	21.63
2008	NM	27.92	2008	WV	25.57
2009	NM	24.03	2009	NV	22.25
2010	WV	27.71	2010	NM	26.53
2011	WV	34.55	2011	NM	29.89
2012	WV	30.64	2012	NM	26.61
2013	WV	31.94	2013	NM	24.93
2014	WV	34.90	2014	NM	28.12
2015	WV	39.00	2015	RI	33.03
2016	DC	48.41	2016	WV	48.26
2017	WV	56.07	2017	DC	53.54
2018	WV	48.88	2018	DC	45.58
2019	DC	55.87	2019	WV	48.09
2020	DC	75.64	2020	WV	74.47
2021	WV	84.40	2021	DC	81.53
2022	DC	87.98	2022	WV	74.74
2023	DC	96.71	2023	WV	79.02

Figure 2: States with first or second-highest overdose rates in each year.

Figure 3 is a plot of the observed overdose death rate across the contiguous US and DC.

Counties that are gray are suppressed per the data use agreement, except for the state of Connecticut see section 2.3. There is a clear concentration of high overdose rates in WV and in the north-eastern counties of Kentucky. There also appears to be a concentration of deaths in New Mexico. Other than those regions, the overdose rate in the USA is pretty much the same regardless of location.

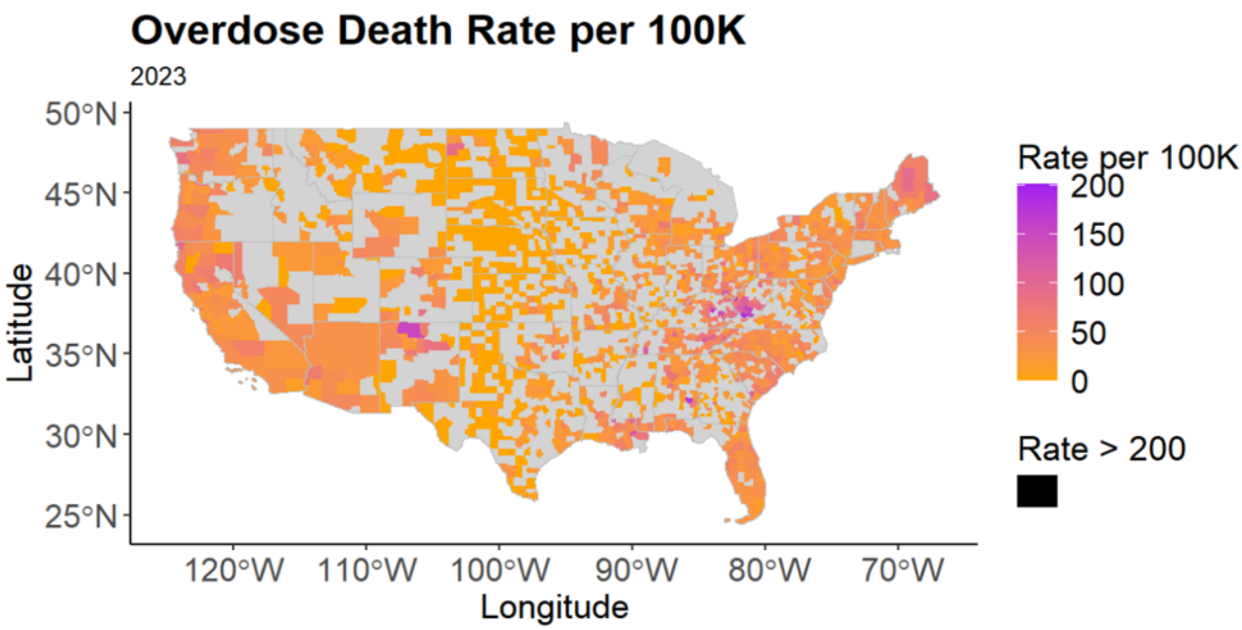


Figure 3: Observed overdose death rate per 100k in USA

Moran's I statistic is a way to test for spatial correlation. A Moran's I of 0 means no spatial correlation, < 0 means negative spatial correlation, > 0 means positive spatial correlation. I look at Moran's I in each year at the county and state level, see Figure 4. The p.values are from a hypothesis test where the null is that there is no spatial correlation or negative correlation, the alternative is that there is positive correlation. So, a significant p.value implies that there is positive spatial correlation. To ensure the same neighbors list and weights matrix is used for all years at the county level and at the state level, the following was done. At the state level, Moran's I is calculated on the contiguous states in 2010-2023. However, at the county level, Moran's I is calculated on all contiguous states except Connecticut, in 2010 to 2023. Connecticut is not included at the county level for any years because of the boundary changes in 2022. The years 2005 to 2009 are not included because the population estimates are suppressed for some counties by ACS. Looking at Figure 4, Moran's I at the county level for all years implies positive spatial correlation. However, at the state level Moran's I implies spatial correlation only for the years 2010, 2011 and 2015 to 2019. It appears that the state level smooths the spatial relationship. But the county level is granular enough that the spatial relationship is significant.

County Level			State Level		
Year	Moran's I	P.value	Year	Moran's I	P.value
2010	0.30	<0.001	2010	0.04	0.25
2011	0.31	<0.001	2011	0.06	0.18
2012	0.27	<0.001	2012	0.15	0.04
2013	0.20	<0.001	2013	0.11	0.09
2014	0.08	<0.001	2014	0.11	0.08
2015	0.22	<0.001	2015	0.24	<0.05
2016	0.28	<0.001	2016	0.25	<0.05
2017	0.31	<0.001	2017	0.30	<0.001
2018	0.31	<0.001	2018	0.30	<0.001
2019	0.26	<0.001	2019	0.15	<0.05
2020	0.37	<0.001	2020	0.12	0.06
2021	0.42	<0.001	2021	0.08	0.14
2022	0.38	<0.001	2022	0.07	0.17
2023	0.33	<0.001	2023	0.06	0.19

Figure 4: Moran's I at the county and state level by year.

At the county level, the percent change ranges from a 100% decrease to a 3913.15% increase with a mean increase of 14.18%. At the state level, the percent change ranges from a 66.64% decrease to a 323.27% increase, with a mean increase of 5.53%. For the most part, drug overdoses are increasing from year to year. For a specific state, the percent change is generally the same, but states differ from one another. At the national level, the percent change ranges from a 44.82% decrease to a 28.67% increase with a mean increase of 3.85%. At all spatial levels of percent change (county, state, national), the largest percent changes generally occur in the early years of the dataset; before 2010. This might be because in 2010 the overdose rate is calculated with ACS 5-year estimates, and so there is no missing data anymore. From 2010 onwards the percent change is pretty steady, until 2020 where there is a dramatic increase in the percent change but then stabilizes. Figure 5 is a table of the percent change at the state level, Figure 6 includes a line plot of the percent change at the state and national level. Figure 5 shows that there is quite a bit of fluctuation in the percent change over time even at the national level. With the highest percent change in 2020, at 28.67%. Figure 6 includes a line plot of the percent change for each state and the national change.

Year	Deaths	Population	Rate	Change
2005	58580	286498255.00	20.45	NA
2006	33562	297442934.00	11.28	-44.82
2007	36103	299654293.00	12.05	6.78
2008	37456	302085237.00	12.40	2.91
2009	37979	305012905.00	12.45	0.42
2010	39498	301940492.00	13.08	5.06
2011	42701	304556515.00	14.02	7.18
2012	42941	307064842.00	13.98	-0.26
2013	45355	309439980.00	14.66	4.81
2014	48704	311986080.00	15.61	6.51
2015	54218	314375347.00	17.25	10.48
2016	65023	316407634.00	20.55	19.16
2017	71450	318844184.00	22.41	9.04
2018	68773	320742485.00	21.44	-4.32
2019	72005	322538633.00	22.32	4.12
2020	93186	324412244.00	28.72	28.67
2021	107927	327536032.00	32.95	14.71
2022	110045	328912183.00	33.46	1.54
2023	107454	330207934.00	32.54	-2.74

Figure 5: Percent change from year to year at the national level.

Figure 6 includes a line plot of the percent change for each state and the national change as well as the percentage of deaths belonging to certain demographic categories. The "Percent Change of the Overdose Rate" shows that the increase in overdose rate by state over the years, has the occasional increase but then levels off. Looking at the other three plots we can see that the majority of overdoses are male, high school graduate and below, and between the ages of 25 and 64 years old. The line plots appear to be constant, meaning that the percentages do not change over time. Implying that the characteristics of people who die from an overdose in any given year are similar to any other given year.

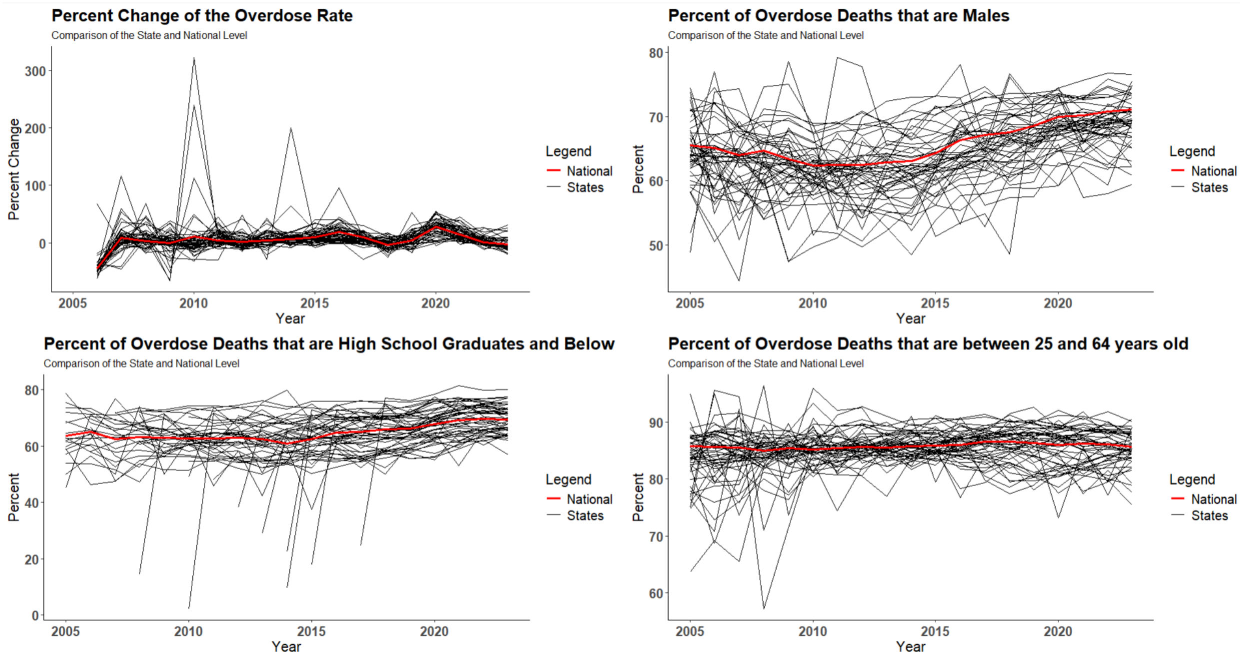


Figure 6: Percent of overdose death by various demographic information.

Figure 7 is a table of the correlation between the overdose rate and covariate data from ACS, see section 2.2. No covariate has strong correlation with the overdose rate. The covariates cognitive disability (Cognitive), self-care disability (SelfCare), insured (Insured), high school education and below (Education), and poverty (Poverty), all exhibit high correlation with each other.

Correlation of Covariates and Rate								
	rate	Cognitive	SelfCare	Insured	Education	Poverty	maritalStatus_D_acs	medianIncome
rate	1.00	0.20	0.18	0.15	0.16	0.15	0.21	0.01
Cognitive	0.20	1.00	0.98	0.97	0.97	0.95	-0.07	0.21
SelfCare	0.18	0.98	1.00	0.97	0.98	0.97	-0.07	0.19
Insured	0.15	0.97	0.97	1.00	0.97	0.93	-0.12	0.28
Education	0.16	0.97	0.98	0.97	1.00	0.97	-0.08	0.20
Poverty	0.15	0.95	0.97	0.93	0.97	1.00	-0.07	0.12
maritalStatus_D_acs	0.21	-0.07	-0.07	-0.12	-0.08	-0.07	1.00	-0.32
medianIncome	0.01	0.21	0.19	0.28	0.20	0.12	-0.32	1.00

Figure 7: Correlation table for all regions.

3 Analysis

I ran a series of models for just West Virginia and then scaled the GPBoost models up to the national level. I chose West Virginia because it has the highest overdose mortality rate of any state in the country. I explored models that captured spatial or temporal effects, fixed and/or random effects. For models that include fixed effects, I used the ACS data as the covariates to determine if they have predictive power regarding the overdose death rate. To compare the fit of various models I used Root Mean Square Error (RMSE) and Mean Absolute Error (MAE).

I ran a series of increasingly complex models in West Virginia at the county level. The models ranged from basic spatial or temporal models, like random walk with drift and SAR, to GPBoost where mixed effects are incorporated with a Gaussian process for space or time. Figure 8 is a table of the counties in the 90th percentile of the overdose rate in 2023, with the average in the whole state. These counties are in the southern part of the state. I am going to compare this to the predicted 90th percentile from each model's predictions for 2023. I will also be comparing training set RMSE and test set RMSE (where applicable) between models. All models are trained on data from the years 2015 to 2022, and if the models design allows for it, it is tested on 2023 data.

FIPS	County	rate
54047	mcdowell	190.08
54011	cabell	178.99
54081	raleigh	154.75
54055	mercer	152.38
54109	wyoming	124.12
54045	logan	119.40
State average		62.78

Figure 8: Counties with an overdose rate in the 90th percentile of all observed rates.

3.1 Random Walk with Drift in West Virginia

To serve as a benchmark, I ran a separate random walk with drift model for each county in West Virginia. Each model was trained on a single county at a time, in the years 2015 to 2022. Then each model was tested in 2023. I calculated the training set and test set RMSE from the errors from all models at once and got a training RMSE of 24.78 and a test RMSE of 22.71. Figure 9 is a table of the predicted 90th percentile of counties, Figure 8 is the observed 90th percentile. Comparing the two tables, the predicted table includes all of the observed counties except for Mercer, and instead includes Kanawha. The order of the predicted counties in the 90th percentile is different from the order of the observed counties in the 90th percentile. The predicted values are lower than the observed values. The average of the predicted rates is 52.73 which is smaller than the observed average rate, 62.78.

FIPS	County	predict
54011	cabell	159.88
54045	logan	155.14
54081	raleigh	143.63
54047	mcdowell	135.72
54039	kanawha	113.89
54109	wyoming	91.19
State average		52.73

Figure 9: Predicted counties with an overdose rate in the 90th percentile.

3.2 Spatial Autoregression in West Virginia

I trained two different SAR models in West Virginia using 2015 to 2022 data, in an attempt to account for spatial effects. The first model was an intercept only model, the second one included various covariates. These two types of SAR models are fit to West Virginia one year at a time. Since SAR is incapable of forecasting to a test set of data, the RMSE I calculated for both models is comparable to the training set RMSEs from the other models. Due to limitations in the model design there will be no test set to compare to. Recall that from section 2.4 there was positive spatial correlation at the county level for the whole USA. Additionally, in West Virginia at the county level there is positive spatial correlation. The Moran's I statistic in West Virginia for every year is different from zero and all have a significant p.value ($< .05$), which implies there is positive spatial correlation.

I calculated the RMSE for the intercept only model from the errors from all models at once, and got an RMSE of 28.63. Based on RMSE, this model performs worse than the benchmark model, indicating that only accounting for spatial effects does not contribute to model accuracy.

The second SAR model includes the covariates cognitive disability (Cognitive), self-care disability (selfCare), health care insured (Insured), poverty (Poverty), education at high school or below (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). Not all covariates had significant p.values for all years. For the model trained on 2015 data, poverty (Poverty) was the only predictor with a significant p.value ($< .05$). For 2016 only cognitive (Cognitive) had a significant p.value. For 2017 self-care (selfCare), insured (Insured), and poverty (Poverty) had significant p.values. For 2018 self-care (selfCare) and education (Education) had significant p.values. For 2019 self-care (selfCare) had a significant p.value. For 2020 median income (medianIncome) had a significant p.value. For 2021 insured (Insured), poverty (Poverty), and median income (medianIncome) had significant p.values. For 2022 poverty (Poverty) had a significant p.value. The poverty estimate (Poverty) was the most frequently occurring significant predictor. The divorced rate (maritalStatus_D_ACS) was never significant for any year. To compare this SAR model to the previous one and to the benchmark, I calculated a RMSE of 20.03. This model has a smaller RMSE than the intercept only SAR model and the benchmark model, implying that it is a better model. It appears that simply accounting for spatial effects is not enough, it is advantageous to include covariate information. However, without a test set RMSE to compare with as well, this comparison to the benchmark is incomplete.

3.3 Random Forest in West Virginia

I fit two non-parametric models, random forest. The first one I fit was aimed at capturing the temporal effects, and the second one was aimed at capturing the spatial effects. Both types of random forest models were trained on data in 2015 to 2022. But only the temporal one could predict into a test set of 2023 data, as the spatial one is not capable of predicting into the test set.

For the first random forest model, it is actually a different model for each county. Each random forest model was trained on a single county at a time, in the years 2015 to 2022. Then each model was tested in 2023. I chose a lag of 2, meaning each year is predicted based on the previous two years of data. I calculated the training set and test set RMSE from the errors from all models at once. I got a training set RMSE of 24.42 and a test set RMSE of 26.6. Figure 10 is a plot of the observed overdose rate vs the predicted values from the random forest model for the test set. This random forest model has a slightly better training set RMSE than the benchmark, but a worse test set RMSE than the benchmark. Overall the random forest model performs similarly to the benchmark, implying that the increased complexity and lack of parameterization of the random forest model failed to improve on the simpler benchmark model.

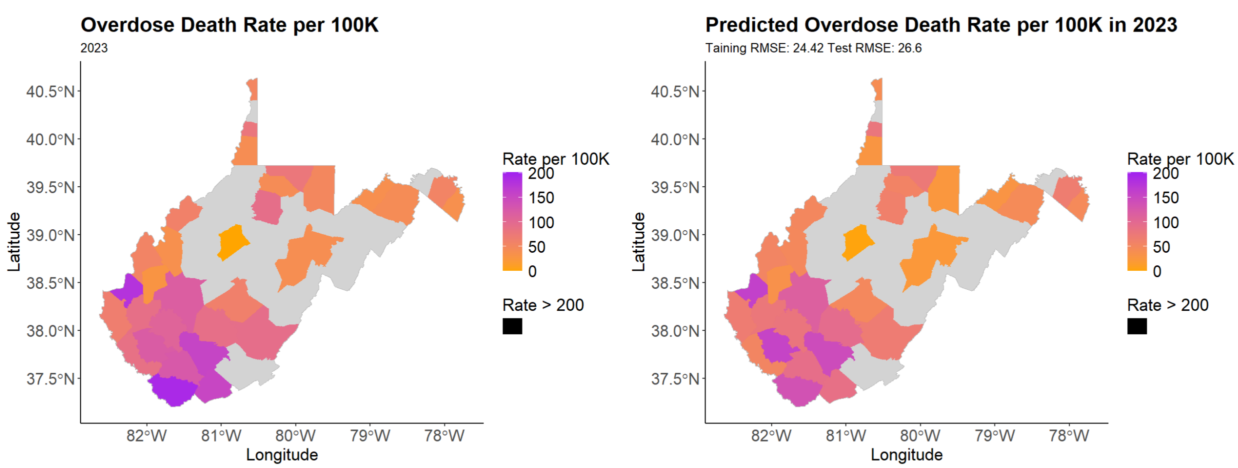


Figure 10: Observed vs predicted values from random forest model.

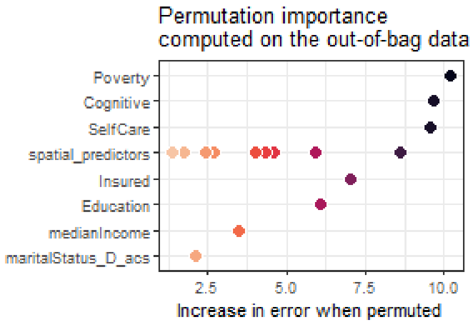
Figure 11 is a table of the predicted counties in the 90th percentile from the random forest model for the test set, 2023 data. The order of the predicted counties in the 90th percentile is different than the observed (Figure 8), and does not include Mercer or Wyoming, but instead Kanawha and Fayette. The predicted values are lower than the observed values.

FIPS	County	predict
54011	cabell	159.88
54045	logan	155.14
54081	raleigh	143.63
54047	mcdowell	135.72
54039	kanawha	113.89
54019	fayette	95.07
State average		57.50

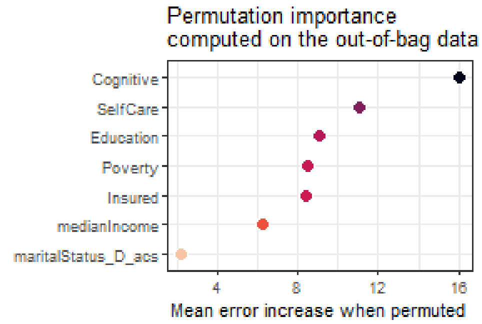
Figure 11: Predicted counties with an overdose rate in the 90th percentile.

The second random forest model aimed to account for the spatial effects. I fit a different random forest model to each year in 2015 to 2022. I included the following covariates: cognitive (Cognitive), self-care (selfCare), insured (Insured), poverty (Poverty), education (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). I then calculated the RMSE from all models at once and got 11.75, this was comparable to the training set RMSEs from the other models. But there is no test set RMSE to compare with this model. The training set RMSE is smaller than the training set RMSE for the benchmark model and the previous random forest model. Implying that capturing the spatial effects does better for model accuracy than capturing the temporal effects. This comparison is however limited because I cannot compare test set accuracies.

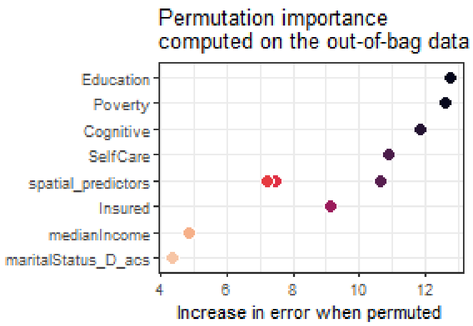
Figure 12 includes plots of the estimated importance of each covariate for each model. The higher the value, the more important the predictor is for improving model accuracy. Cognitive disability (Cognitive) was the most commonly occurring predictor with the highest importance, Figures 12b, 12d, 12f.



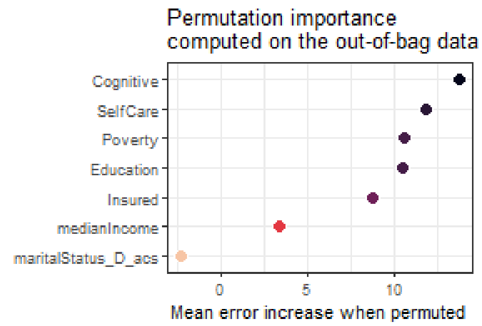
(a) Importance for 2015.



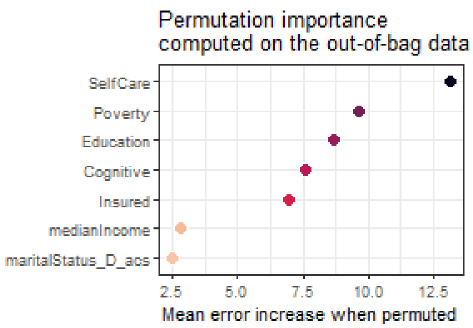
(b) Importance for 2016.



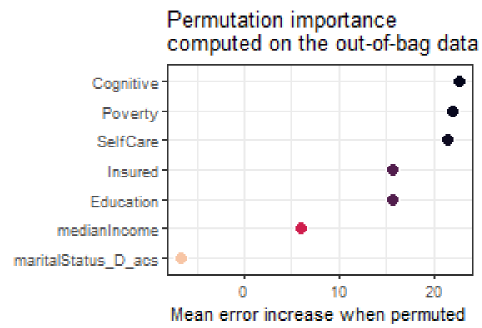
(c) Importance for 2017.



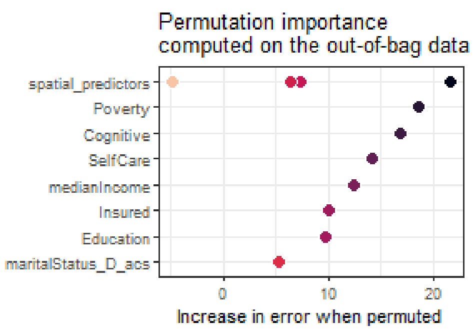
(d) Importance for 2018.



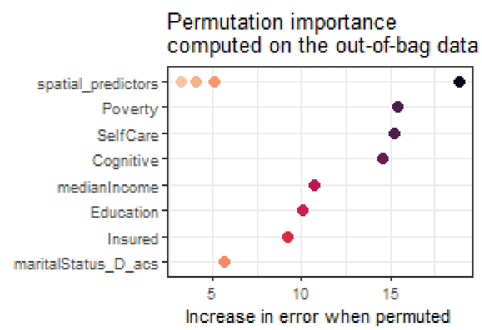
(e) Importance for 2019.



(f) Importance for 2020.



(g) Importance for 2021.



(h) Importance for 2022.

Figure 12: Importance of Covariates.

3.4 Gaussian Process Boosting in West Virginia

To consider spatial or temporal effects with mixed effects I used GPBoost models. The GPBoost models in this section were trained on data in the years 2015 to 2022 and then tested on data in 2023 only for the state of West Virginia. Both of these models had a random intercept by county and include the same fixed effects as the previous models: cognitive (Cognitive), self-care (selfCare), insured (Insured), poverty (Poverty), education (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). Both models were fit with a Gaussian likelihood and an exponential covariance function. The first model I fit included a Gaussian process for space, and random effects for county and for year. The second model included a single Gaussian process for time, shared across all counties, with a random effect for each county.

The first GPBoost model I fit included a spatial Gaussian process for every county in the years 2015 to 2022. The model included random grouped effects (intercept) for each county and random effects (slope) for each Year. With fixed effects cognitive (Cognitive), self-care (selfCare), insured (Insured), poverty (Poverty), education (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). I ran a grid search to choose the best tuning parameters for boosting, as recommended by Sigrist (Sigrist, 2023a). The best tuning parameters from this approach were learning_rate = 1, max_depth = 10, num_leaves = 2^{10}, min_data_in_leaf = 10, lambda_l2 = 0, and nrounds = 880, so those are the settings I used. This model reported a training set RMSE of 9.9 and a test set RMSE of 24.67. Figure 13 is a plot of the observed overdose death rate vs the predicted values from this model for the test set. This model had a much smaller training set RMSE than the benchmark model, but a slightly larger test set RMSE than the benchmark model. The difference between the training set RMSE and the test set RMSE is quite large, implying that this model is over fitting to the training set.

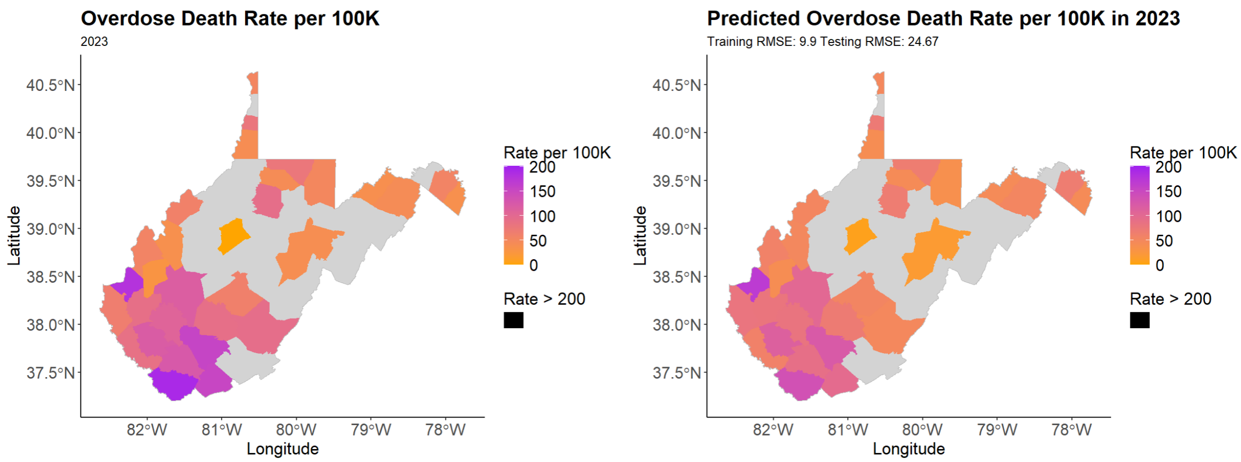


Figure 13: Observed vs predicted values from GPBoost with spatial Gaussian process.

To gauge the contribution of including each fixed effect in improving model accuracy I chose to focus on gain. Figure 14 includes the gain of each fixed effect. A higher gain implies that the corresponding covariate contributed more to improving model accuracy. The estimate of median household income (medianIncome) has the highest gain, followed by self-care (SelfCare), implying they contributed to the model more than the other covariates.

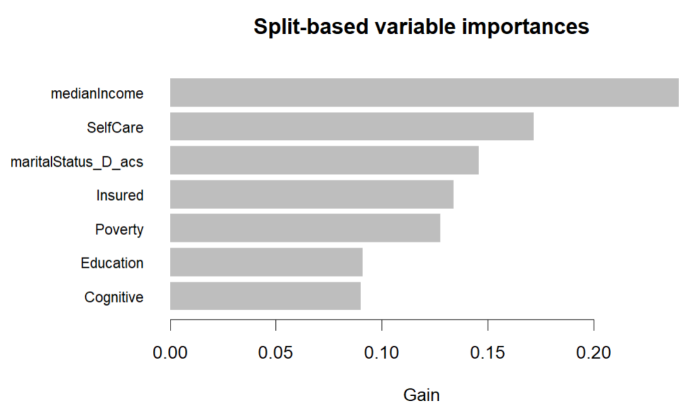


Figure 14: Gain for fixed effects from the GPBoost model with a spatial Gaussian process.

The variance of the error term in 111.42. The variance of the random intercept by county is 0.77. The variance of the random slope for Year is .0001. The spatial variance is 643.70. The variance of the overdose rate in WV in 2015 to 2022 is 1224.66. The variance of the spatial effects is larger than the random effects, implying that it is explaining more of the variability in the overdose rate. The variance for the random slope and random intercept are very small, implying that including these random grouped effects is not contributing to model accuracy. The remaining variance not explained here is attributed to the fixed effects.

Figure 15 is a table of the predicted counties in the 90th percentile for the test set, 2023 data. The order of the predicted counties in the 90th percentile is different than the observed (Figure 8), and does not include Wyoming, but instead Kanawha. The predicted values are lower than the observed.

FIPS	County	pred
54011	cabell	168.44
54047	mcdowell	136.28
54081	raleigh	122.23
54045	logan	113.36
54039	kanawha	101.52
54055	mercer	95.96
State average		52.59

Figure 15: Predicted counties with an overdose rate in the 90th percentile.

The second GPBoost model included a single temporal Gaussian process for every county in the years 2015 to 2022. The model included random effects (intercept) for each county, and fixed effects cognitive (Cognitive), self-care (selfCare), insured (Insured), poverty (Poverty), education (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). I ran a grid search to choose the best tuning parameters for boosting, as recommended by Sigrist (Sigrist, 2023a). The best tuning parameters from this approach are learning_rate = 1, max_depth = 10, num_leaves = 2^{10}, min_data_in_leaf = 10, lambda_l2 = 10, and nrounds = 90, so those are the settings I used. This model reported a training set RMSE of 16.26 and a test set RMSE of 27.14. Figure 16 is a plot of the observed overdose death rate vs the predicted values from this model. This model reported a smaller training set RMSE than the benchmark model but a larger test set RMSE than the benchmark model. Finally, the difference between the training set RMSE and the test set RMSE suggests this model may be somewhat over fit to the training set.

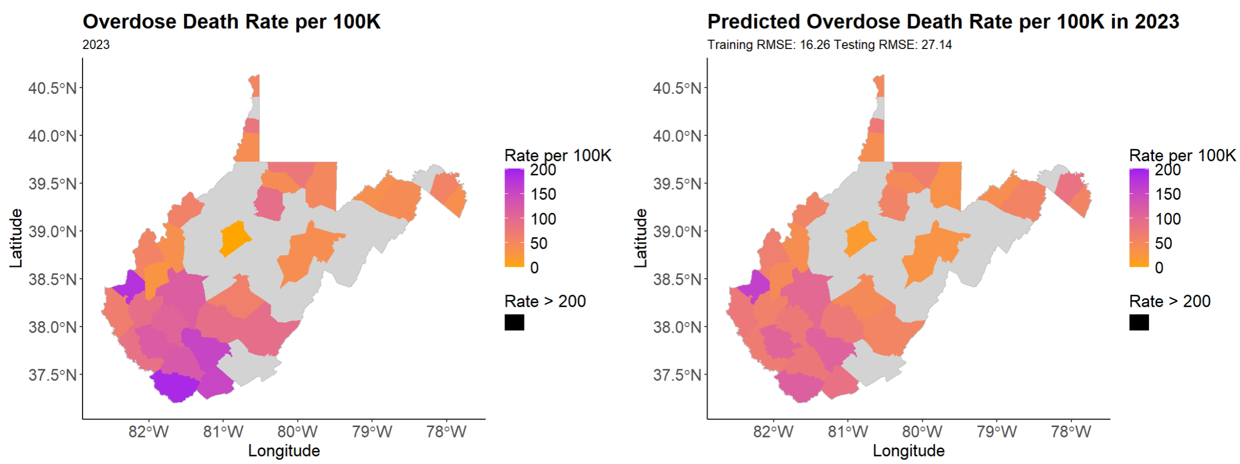


Figure 16: Observed vs predicted values from GPBoost model with a temporal Gaussian process.

Gain measures the contribution of including each fixed effect in improving model accuracy. Figure 17 includes the gain of each fixed effect. A higher gain implies that the corresponding covariate contributed more to improving model accuracy. The divorced rate (maritalStatus_D_ACS) has the highest gain, followed by self-care (SelfCare), implying they contributed to the model more than the other covariates.

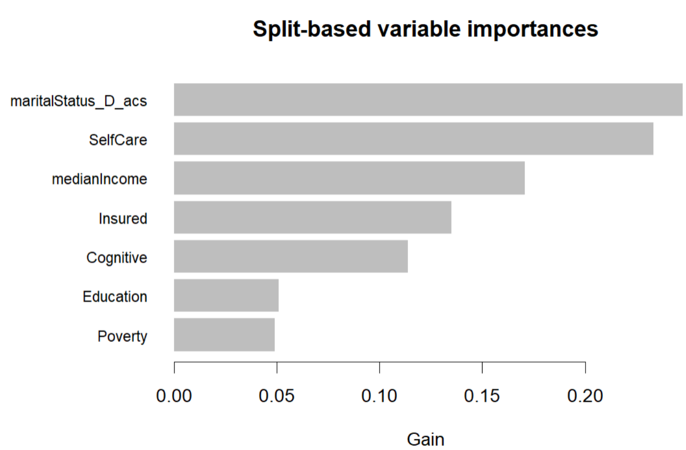


Figure 17: Gain for fixed effects from the GPBoost model with a temporal Gaussian process.

The variance of the error term is 305.03. The variance of the random effect is 749.62. The temporal variance is 179.21. The variance of the overdose rate in WV in 2015 to 2022, is 1224.66. The variance of the random effects is larger than the other effects implying that it is explaining more of the variability in the overdose rate than the temporal effect. The remaining variance is attributed to the fixed effects, which may not be much as the error variance is quite large.

Figure 18 is a table of the predicted counties in the 90th percentile for the test set, 2023 data. The order of the predicted counties in the 90th percentile is different than the observed (Figure 8), and does not include Wyoming, but instead Kanawha. The predicted values are lower than the observed. The order of the predicted counties in the 90th percentile are the same as the previous GPBoost model (Figure 15), and the predicted values are very similar as well.

FIPS	County	pred
54011	cabell	160.05
54047	mcdowell	114.29
54081	raleigh	110.03
54045	logan	106.37
54039	kanawha	99.17
54055	mercer	85.38
State average		53.96

Figure 18: Predicted counties with an overdose rate in the 90th percentile.

This model had larger training and test set RMSEs than the previous GPBoost model. However the difference between this model's training and test set RMSE is smaller than the previous GPBoost model, implying that this model is not over fitting to the training set as much. The fixed effect with the highest gain for the temporal model was the divorced rate (maritalStatus_D_ACS), but for the spatial process it was household median income (medianIncome). Interestingly, the next highest gain for both models was for self-care disability (SelfCare). The first GPBoost model had smaller error variance, 111.42, than the second model, 305.03. The first model primarily attributed its ability to explain the variability in the response to the spatial effects, with a variance of 643.70, but the second model attributed it mostly to the random effect, with a variance of 749.62. The second model is simpler and runs faster than the first GPBoost model.

3.5 Summary of Models fit in West Virginia

For West Virginia I fit a random walk with drift to serve as a benchmark, a SAR model, a random forest to account for time, a random forest to account for space, GPBoost with a Gaussian process for space, and GPBoost with a Gaussian process for time. Figure 19 is a table of the training set and test set RMSE from each model fit in West Virginia. The GPBoost model with the spatial process has the smallest training set RMSE, 9.9. The benchmark model, the simplest model I fit, has the smallest test set RMSE, 22.71. Also the benchmark model has the smallest difference between the training and test set RMSEs. Implying that it does a good job of not over or under fitting to the training set. Whereas the GPBoost model with the spatial process has the largest difference between the training and test set RMSEs implying that it is over fit to the training set, making it poor at predicting into a test set.

Model	Train RMSE (2015-2022)	Test RMSE (2023)	Train MAE (2015-2022)	Test MAE (2023)
Random Walk with Drift	24.83	22.71	19.33	17.21
SAR intercept only	28.63	-	20.27	-
SAR with Covariates	20.03	-	15.51	-
Random Forest	24.42	26.60	17.89	18.98
Random Forest, Space	11.75	-	7.98	-
GPBoost, Space	9.90	24.67	7.30	17.88
GPBoost, Time	16.26	27.14	11.97	20.28

Figure 19: Comparison of models run in West Virginia.

3.6 Gaussian Process Boosting in the USA

Next I decided to scale up and run both the spatial and the temporal GPBoost models previously fit to West Virginia, to the whole USA. This however proved to be more computationally complex than what my system could handle. So, I divided the USA into 8 regions and trained one model for each region. Each model is trained in the years 2015 to 2022 and then tested in 2023. The 8 regions are as follows:

- Northeast Region: Maine, Vermont, New Hampshire, Massachusetts, Rhode Island, Connecticut, New York, New Jersey, Pennsylvania, Delaware, Maryland, and The District of Columbia.
- Southeast Region: Georgia, North Carolina, South Carolina, and Virginia.
- Appalachian Region: Arkansas, Tennessee, West Virginia, and Kentucky.
- Gulf Coast Region: Florida, Alabama, Mississippi, and Louisiana.
- Southwest Region: Arizona, New Mexico, Oklahoma, and Texas.
- Great Lakes Region: Wisconsin, Michigan, Ohio, Illinois, and Indiana.
- Midwest Region: North Dakota, South Dakota, Nebraska, Kansas, Minnesota, Iowa, and Missouri.
- Western Region: Washington, Montana, Idaho, Wyoming, Colorado, Utah, Nevada, California, and Oregon.

The first GPBoost model includes a Gaussian process for space, random effects for county and year and various fixed effect; following the same components as the first GPBoost model fit to West Virginia. To fit the model, I also used the same tuning parameters as was done on West Virginia, except I decreased the number of boosting iterations, nrounds, to 40. This GPBoost model is trained to one of the eight regions discussed above at a time, for the years 2015 to 2022, then tested in 2023. I calculated the training set RMSE from the errors of all the models at once and got 11.62, I did the same thing for the test set and got 18.05. I also calculated the training set MAE to be 7.93 and the test set MAE to be 11.39. The RMSE and MAE for each individual region are listed in Figure 21. A plot of the predicted values from each model can be viewed in Figure 20.

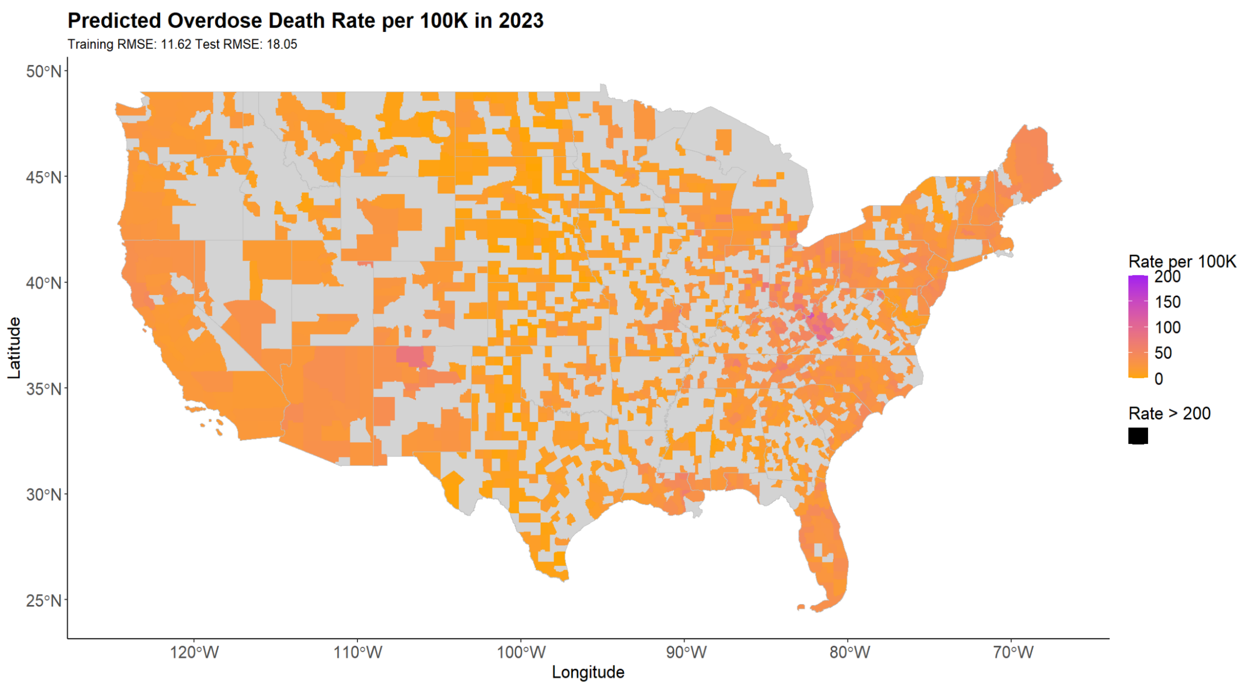


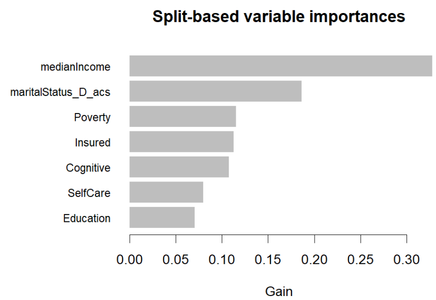
Figure 20: GPBoost model prediction for the 8 regions in the USA.

Figure 21 includes the training set and test set RMSEs and MAEs for each region. As well as the variance of the overdose rate (the response) for the training set data. The training and test set RMSEs seem to vary somewhat from region to region. The Appalachian region has the highest test set RMSE and has the highest variance of the overdose rate. Perhaps this is related to the fact that within the Appalachian region is West Virginia which is known for having the highest overdose rates in the country, but the neighboring regions have similar overdose rates to that of the rest of the US, which is lower. The West region has similar training set and test set RMSEs and MAEs to that of the Appalachian region, despite this it actually has a much smaller response variance.

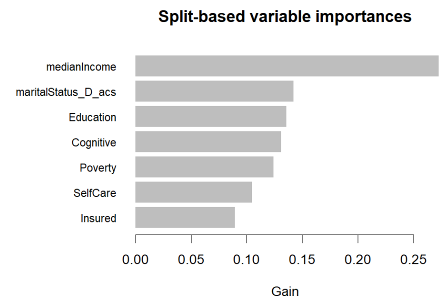
Region	Train RMSE	Test RMSE	Train MAE	Test MAE	Response Variance
Northeast	8.16	17.15	5.92	12.16	251.95
Southeast	11.98	16.02	8.64	11.14	322.90
Appalachian	15.78	25.27	11.49	16.20	617.31
Gulf Coast	9.79	19.24	7.10	11.50	242.09
Southwest	10.43	17.00	7.08	10.62	223.71
Great Lakes	9.12	11.57	6.53	8.18	268.50
Midwest	10.51	13.97	7.18	9.04	194.83
West	14.94	24.28	9.53	15.09	322.76
National	11.62	18.05	7.93	11.39	533.03

Figure 21: GPBoost model accuracy for the 8 regions in USA.

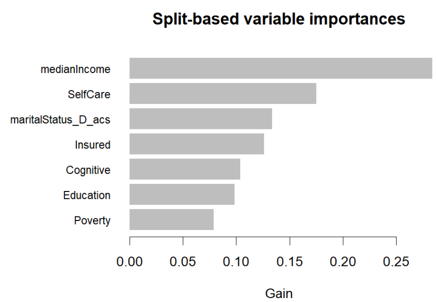
Figure 22 includes the gain of each fixed effects for each of the 8 models. Median household income (medianIncome) has the highest gain for the Northeast, Southeast, Appalachian, Gulf Coast, Southwest, Midwest, and Great Lakes regions; Figures 22a, 22b, 22c, 22d, 22e, 22g and 22f. The poverty estimate (Poverty) has the highest gain in the West region, Figure 22h. Taking into account the gain of all fixed effects for all models at once, it appears that median income (medianIncome) contributes the most to explaining the variability in the overdose rate.



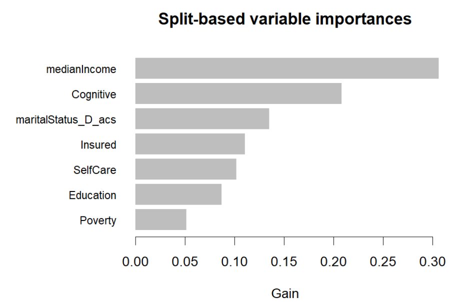
(a) Northeast



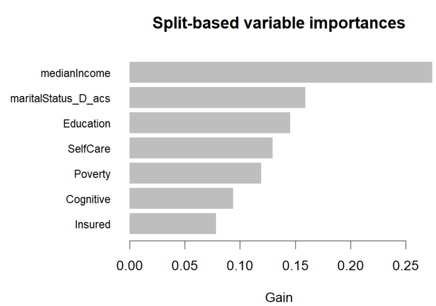
(b) Southeast



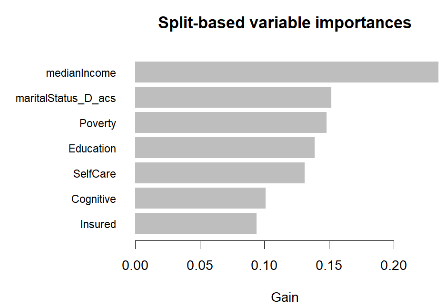
(c) Appalachian



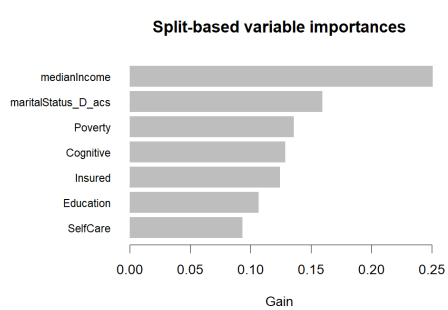
(d) Gulf Coast



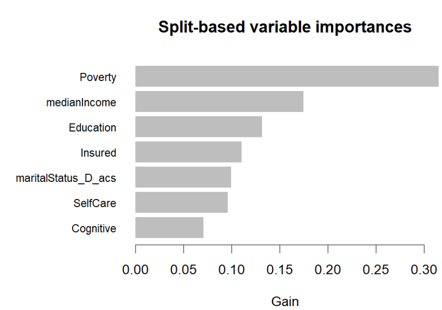
(e) Southwest



(f) Great Lakes



(g) Midwest



(h) West

Figure 22: Gain of the fixed effects for each region.

Figure 23 is a table of the variance attributed to error, random effects, and spatial effects. The last column is the variance of the response for the training set data used in each region. When the variance of the error term is small compared to the variance of the response, this implies that the model does well in capturing the total variance of the response. Since the variance of the error term is smallest in the Northeast region, it implies that this model captures the variance in the overdose rate well. Whereas the largest variance in the error term occurs in the West region, implying this model does not capture the variance in the overdose rate well. The variance of the spatial effect is larger than the random effects for all 8 regions. This implies that the spatial effects are explaining more of the variability in the overdose rate than the random effects. The remaining variance captured by the model is attributed to the fixed effects. Since the error variance is small compared to the response variance, and the random effects variance is large, there does not seem to be much leftover variance explained by the fixed effects.

Covariance Parameters (random effects):					Response
Region	Error Term	Random Intercept	Random Slope	Gaussian Process	Variance
Northeast	75.21	3.24	0.00	99.96	251.95
Southeast	160.38	48.46	0.00	52.17	322.90
Appalachian	276.32	19.03	0.00	252.42	617.31
Gulf Coast	106.41	9.22	0.00	106.40	242.09
Southwest	119.40	5.51	0.00	149.10	223.71
Great Lakes	93.70	16.35	0.00	89.61	268.50
Midwest	121.24	18.25	0.00	39.03	194.83
West	240.47	17.60	0.00	62.00	322.76

Figure 23: GPBoost variance for random effects.

Figure 24 is a summary of the observed values, the predicted values, and the residuals. The predicted values for the training set and test set data are both underestimating the extremes of the observed data for each set. The average of the predicted values in the training set is 18.3, which is the same as the observed average. But the highest predicted value for the training set, 134.3, is smaller than the observed overdose rate, 266.31. The average of the predicted values for the test set, 18.88, is smaller than the observed average of 24.05. The predicted high, 133.52, is much smaller than the observed high, 212.99.

Summary Statistics of GPBoost Predictions and Residuals						
	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Training Set Rate	0.00	5.70	14.61	18.30	25.61	266.31
Training Set Prediction	-0.49	9.68	15.61	18.30	23.66	134.30
Training Set Residuals	-55.54	-6.41	-1.79	0.00	4.50	201.21
Test Set Rate	0.00	8.80	19.34	24.05	32.87	212.99
Test Set Prediction	-1.10	9.97	16.13	18.88	24.65	133.52
Test Set Residuals	-56.23	-4.89	1.95	5.17	11.26	201.88

Figure 24: Summary of GPBoost predictions.

The next GPBoost included a single temporal Gaussian process across all counties, a random effect by county, and various fixed effects. This is the same model setup as was previously done on West Virginia, including the same tuning parameters. This GPBoost model is trained to one of the eight regions discussed at the beginning of this section, for the years 2015 to 2022, then tested in 2023. I calculated the training set RMSE from the errors of all the models at once and got 10.51, I did the same thing for the test set and got 17.41. I also calculated the training set MAE to be 7.05 and the test set MAE to be 11.43. This GPBoost model has smaller training and test set RMSEs than the previous GPBoost model which included a Gaussian process for spatial effects. The RMSE and MAE for each individual region are listed in Figure 26. A plot of the predicted values from each model can be viewed in Figure 25.

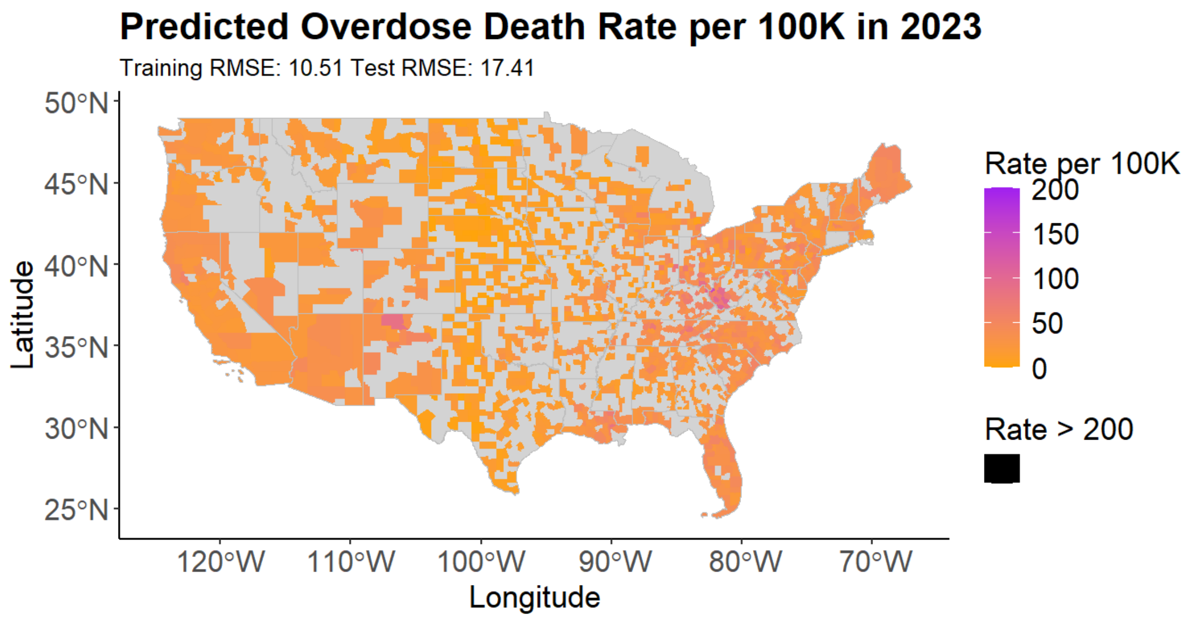


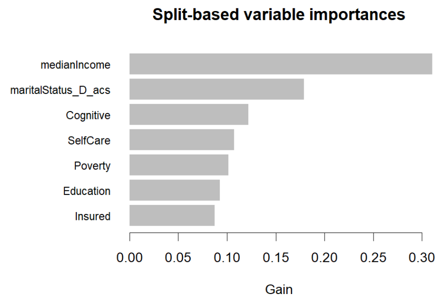
Figure 25: GPBoost model prediction for the 8 regions in USA.

Figure 26 includes the training set and test set RMSEs and MAEs for each region. As well as the variance of the overdose rate for the training set data. The training and test set RMSEs seem to vary somewhat from region to region. Again, the Appalachian region has the highest test set RMSE and has the highest variance of the overdose rate. The West region has similar training set and test set RMSEs and MAEs to that of the Appalachian region, despite this it actually has a much smaller response variance. The training and test set RMSEs by each region for this temporal model are all slightly smaller than for the spatial GPBoost model.

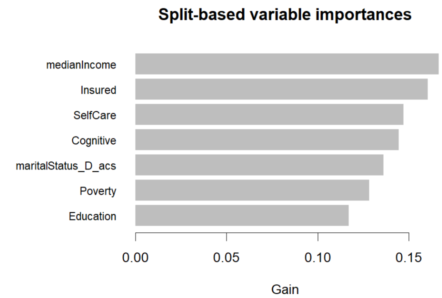
Region	Train RMSE	Test RMSE	Train MAE	Test MAE	Response Variance
Northeast	6.83	20.65	4.95	15.83	251.95
Southeast	9.97	14.80	7.07	10.89	322.90
Appalachian	13.73	23.32	9.97	16.18	617.31
Gulf Coast	7.98	17.16	5.83	10.47	242.09
Southwest	9.72	16.77	6.47	10.49	223.71
Great Lakes	8.31	11.40	5.91	8.30	268.50
Midwest	10.25	13.77	6.99	9.22	194.83
West	14.25	22.93	8.65	13.95	322.76
National	10.51	17.41	7.05	11.43	533.03

Figure 26: GPBoost model accuracy for the 8 regions in USA.

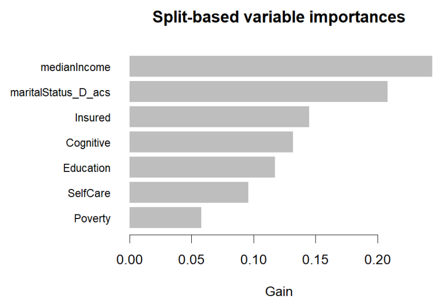
Figure 27 includes the gain of each fixed effects for each of the 8 models. Median household income (medianIncome) has the highest gain for the Northeast, Southeast, Appalachian, Gulf Coast, Southwest, and Midwest regions; Figures 27a, 27b, 27c, 27d, 27e, and 27g. Self-care disability (SelfCare) has the highest gain in the Great Lakes region, Figure 27f. The estimates of people with a high school education and below (Education) had the highest gain in the West region, Figure 27h. Median income (medianIncome) however has the second-highest gain in the Great Lakes and West regions, Figures 27f and 27h. Taking into account the gain of all fixed effects for all models at once, it appears that median income (medianIncome) contributes the most to explaining the variability in the overdose rate. The gain of each covariate for this model are very similar to the gain reported for the previous GPBoost model.



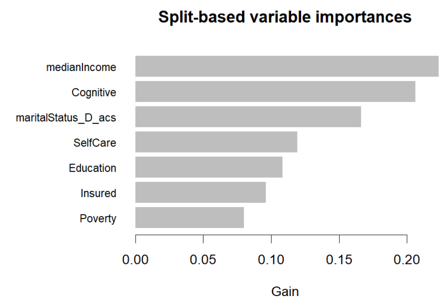
(a) Northeast



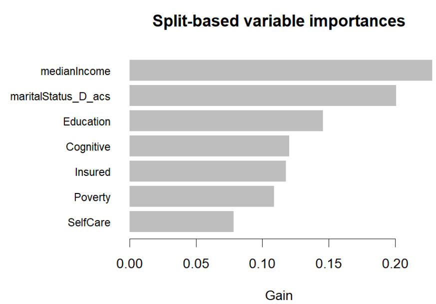
(b) Southeast



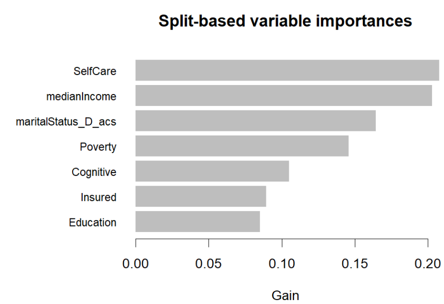
(c) Appalachian



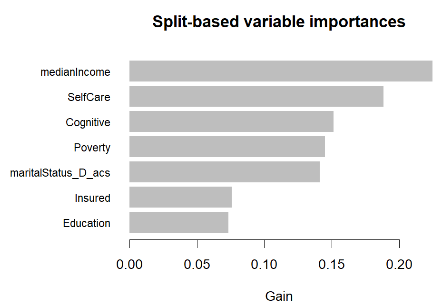
(d) Gulf Coast



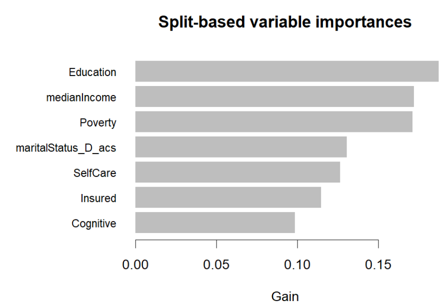
(e) Southwest



(f) Great Lakes



(g) Midwest



(h) West

Figure 27: Gain of the fixed effects for each region.

Figure 28 is a table of the variance attributed to error, random effects, and temporal effects. The last column is the variance of the response for the training set data used in each region. Since the variance of the error term is smallest in the Northeast region, it implies that this model captures the variance in the overdose rate well. Whereas the largest variance in the error term occurs in the West region, implying this model does not capture the variance in the overdose rate well. The variance of the random effects is larger than then temporal effect for all 8 regions. This implies that the random effects are explaining more of the variability in the overdose rate than the temporal effect. This is in contrast to the previous model which reported that the Gaussian process for the spatial effects accounted for most of the variability in the response. The remaining variance captured by the model is attributed to the fixed effects. Since the error variance is small compared to the response variance, and the random effects variance is large, there does not seem to be much leftover variance explained by the fixed effects. The variance of the Error_term is smaller for every region for this model compared to the previous model, Figure 23.

Covariance Parameters (random effects):				Response
Region	Error Term	Random Intercept	Gaussian Process	Variance
Northeast	53.20	153.29	35.64	251.95
Southeast	112.53	147.12	59.91	322.90
Appalachian	213.79	322.80	70.04	617.31
Gulf Coast	72.32	116.44	40.01	242.09
Southwest	106.21	86.57	7.69	223.71
Great Lakes	78.47	155.91	16.61	268.50
Midwest	116.95	57.69	2.75	194.83
West	223.49	71.35	15.72	322.76

Figure 28: GPBoost variance for random effects.

Figure 29 is a summary of the observed values, the predicted values, and the residuals. The predicted values for the training set and test set data are both underestimating the extremes of the observed data for each set. The average of the predicted values in the training set is 18.3, which is the same as the observed average. But the highest predicted value for the training set, 158.32, is smaller than the observed overdose rate, 266.31. The average of the predicted values for the test set, 23.45, is smaller than the observed average of 24.05. The predicted high, 152.91, is much smaller than the observed high, 212.99.

Summary Statistics of GPBoost Predictions and Residuals						
	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Training Set Rate	0.00	5.70	14.61	18.30	25.61	266.31
Training Set Prediction	-4.56	9.08	15.46	18.30	24.14	158.32
Training Set Residuals	-65.55	-5.56	-1.06	0.00	4.05	194.38
Test Set Rate	0.00	8.80	19.34	24.05	32.87	212.99
Test Set Prediction	0.92	13.72	21.28	23.45	29.80	152.91
Test Set Residuals	-67.50	-8.84	-2.14	0.60	6.62	202.42

Figure 29: Summary of GPBoost predictions.

The GPBoost model that included a temporal Gaussian process had a smaller training set and test RMSE set than the GPBoost model that included a spatial Gaussian process, this is for the RMSEs by each region and pooled. The temporal model additionally had smaller error variance terms for all regions. This implies that the GPBoost model with the temporal Gaussian process did better capturing the full variability of the response. The temporal Gaussian model also ran significantly faster than the GPBoost model with the spatial process. Both models overwhelmingly reported that the fixed effect for median household income (medianIncome) is the most important of the fixed effects. The Appalachian region had the highest training and test set RMSEs for both GPBoost models.

4 Conclusion

To evaluate why drug overdoses have been on the rise over the last two decades, I have run models to capture the potential effects of time, space, and various socioeconomic variables. The socioeconomic variables I decided to investigate were cognitive (Cognitive), self-care (selfCare), insured (Insured), poverty (Poverty), education (Education), divorced rate (maritalStatus_D_ACS), and household median income (medianIncome). I first ran a battery of models on just West Virginia, and then expanded two of those models to the whole USA.

In West Virginia, I ran various models specialized in capturing spatial and temporal effects. Using random walk with drift as a benchmark, the smallest training set RMSE belonged to the GPBoost model with the spatial Gaussian process. However, the test RMSE for this model was much larger. The benchmark model, while not having the smallest training set RMSE did have the smallest test set RMSE. GPBoost did not preform well in terms of forecasting into a test set. Of the covariates I included in the GPBoost model, median household income (medianIncome) had the highest gain. Which demonstrates the potential connection between income and drug overdoses.

Of all the covariates, median household income (medianIncome) and poverty (Poverty) were the most important for the majority of the models. With both being related to income, clearly income is the most related to the overdose rate, compared to the other socioeconomic variables I investigated. For the SAR model, poverty (Poverty) had a significant p.value for 4 of the 8 years, the most of any variable. For the spatial random forest model, cognitive disability (Cognitive) had the highest importance for 3 of the 8 years, with poverty (Poverty) having the highest importance in 1 year and the highest importance after the spatial effects in 2 other years. For the GPBoost model with the spatial Gaussian process, median income (medianIncome) had the highest gain. For the GPBoost model with the temporal Gaussian process, the divorced rate (maritalStatus_D_ACS) had the highest gain.

Next I fit two GPBoost models to 8 regions in the USA. The first GPBoost model included a Gaussian process for space, random effects for county and year, and fixed effects. An analysis of the fixed effects revealed that median income (medianIncome) had the largest gain in 7 of the 8 regions, and had the largest gain after poverty (Poverty) in the remaining region. The second model included a Gaussian process for time, random effects for county and the same fixed effects. Interestingly, this second model preformed better in terms of training and test set RMSE across all regions. An analysis of the fixed effects revealed that median income (medianIncome) had the largest gain in 6 of the 8 regions, and had the second-largest gain in the remaining 2 regions. Further demonstrating the relationship between median income (medianIncome) and the overdose rate.

Future work may benefit from fitting the GPBoost models to all counties across the USA at once, instead of breaking it down by region as was done here. This may improve the accuracy of these models, and better preserve the spatial relationships. Future work may also examine different covariates with different random effects. Or even define the spatial relationships differently than was done here. It also might be beneficial to discover exactly what income brackets or other income related variables are most associated with higher rates of drug overdoses.

5 Software

For this analysis I exclusively used the R programming language within the integrated development environment, RStudio. In this section, I discuss and walk through the code I used to fit all the models discussed in this paper. I have made my code and other necessary files available on GitHub here: https://github.com/PennyRose05/MasterStatisticsProject (Overholt, 2026). For this project I used the restricted data files from NCHS, so instead I have included the public files from CDC WONDER at my GitHub repository.

The file formatCntyData.R is the code I wrote to read and format the restricted data files from NCHS, as well as the code I wrote to retrieve ACS data. The file FunctionsCntyDeath.R contains functions I wrote, as well as the packages needed for data wrangling, and producing plots and tables. The file data_analysis.R contains code I wrote to look at the additional information provided with the restricted NCHS data files, and to look at ACS covariate data. The file randomForestCnty.R contains the code I wrote for the random walk with drift model, the temporal random forest model, and the spatial random forest model. The file SAR_cnty.R contains the code I wrote for the SAR models, the intercept only model and the one that includes the covariates. The file gpboost.R contains the code to run the GPBoost models, the one with a spatial Gaussian process and the one with a temporal Gaussian process, as well as a third one that I did not do much with. The file formatToyData.R contains the code I wrote to format the public mortality data from CDC WONDER so that is it compatible with my existing code. I got the county level shapefiles from the Census Bureau, I used the 2021 files for all years (U.S. Census Bureau, 2021). I chose the year 2021 because that pre-dates Connecticut changing its counties boundaries.

5.1 Random Walk and Random Forest

To run the random walk with drift and random forest models, I used the randomForest and forecast packages (R. Hyndman et al., 2025; R. J. Hyndman & Khandakar, 2008; Liaw & Wiener, 2002).

This code chunk will load the packages and read the data. Then also define some variables to calculate the pooled MSE and MAE later.

```
source("FunctionsCntyDeath.R")
library(randomForest) #randomForest
library(forecast) #accuracy()

set.seed(1)

choose_state = "WV"
data <- getOverdoseData(state=choose_state,choose_year="all",more=
   FALSE)

data <- data[data$Year >= 2015,]
curr_FIPS <- unique(data$FIPS)
info <- matrix(ncol=8,nrow=length(curr_FIPS))

sum_sq_pred <- 0
sum_sq_pred_bench <- 0
n_sum_sq <- 0

sum_abs_pred <- 0
sum_abs_pred_bench <- 0
```

```
par(mfrow=c(5,5))
```

This next chunk of code will loop through every county, and fit a random walk with drift and a random forest model. The random forest model is fit with a lag of 2, and the predictors and response training set data are passed to randomForest() to fit the model. Then the random walk with drift model is fit with naive(). It will also calculate the pooled MSE and MAE.

```
for(i in 1:length(curr_FIPS))
{
  temp <- data[data$FIPS == curr_FIPS[i] & !is.na(data$rate),]
  data_ts <- ts(temp$rate,
                start=c(min(temp$Year),1),frequency=1)

  test_yr=2023
  chooseLag=2

  data_ts_org <- window(data_ts,end = c((test_yr-1),1)) #train with
     2015-2022
  lag_order <- chooseLag
  horizon <- 1 # the forecast horizon
  data_ts_mbd <- embed(data_ts_org, (lag_order + 1))

  y_train <- data_ts_mbd[, 1] # the target
  X_train <- data_ts_mbd[, -1] # everything but the target
  y_test <- window(data_ts, start = c(test_yr,1), end = c(test_yr,1))
  X_test <- data_ts_mbd[nrow(data_ts_mbd), c(1:(lag_order))]
  set.seed(1)
   # fit the model
   fit_rf <- randomForest(X_train, y_train)
   mse <- fit_rf$mse
   trainingRMSE <- sqrt(mse[length(mse)])
   # predict using the test set
   forecasts_rf <- predict(fit_rf, X_test)

    y_fore <- ts(
     forecasts_rf,
     start = c(test_yr, 1),
     frequency = 1
    )

# add the forecasts to the original table

  testacc=accuracy(y_fore, y_test)
  temp <- forecast(naive(data_ts_org), h = horizon)
```

```
  benchmark <- accuracy(temp,y_test) #is a basic model to compare to

  info[i,1] <- curr_FIPS[i]
  info[i,2] <- trainingRMSE
  info[i,3] <- testacc[2] # testing RMSE
  info[i,4] <- y_fore # predict for test year
  info[i,5] <- benchmark[1,2]#training RMSE
  info[i,6] <- benchmark[2,2]#testing RMSE
  info[i,7] <- temp$mean
  info[i,8] <- length(data_ts)

  # plot(data_ts,main=paste(curr_FIPS[i],info[i,3]))+
  #   points(y_fore,col="red",pch=16)

  obs_train <- data_ts_org[3:length(data_ts_org)]
  sum_sq_pred <- sum((obs_train - fit_rf$predicted)^2)+sum_sq_pred
  sum_sq_pred_bench <- sum((data_ts_org[2:length(data_ts_org)] -
     temp$fitted[2:length(temp$fitted)] )^2)+sum_sq_pred_bench
  n_sum_sq <- length(fit_rf$predicted)+n_sum_sq

  sum_abs_pred <- sum(abs(obs_train - fit_rf$predicted))+sum_abs_pred
  sum_abs_pred_bench <- sum(abs(data_ts_org[2:length(data_ts_org)] -
     temp$fitted[2:length(temp$fitted)] ))+sum_abs_pred_bench
}

all_rmse_train <- sqrt(sum_sq_pred/n_sum_sq)
all_rmse_train_bench <- sqrt(sum_sq_pred_bench/n_sum_sq)

all_mae_train <- sum_abs_pred/n_sum_sq
all_mae_train_bench <- sum_abs_pred_bench/n_sum_sq

info <- as.data.frame(info)
names(info) <- c("FIPS","TrainRMSE","TestRMSE","predict","
   benchmark_train","benchmark_test","bench_pred","n")

par(mfrow=c(1,1))
plot(x=factor(info$FIPS),info$TestRMSE)+
  points(x=factor(info$FIPS),info$benchmark_test,col="red")+
  abline(h=c(39.58))

data_test <- data[data$Year == test_yr,]
data_test <- left_join(data_test,info,by=c("FIPS"))
data_test$predict <- as.numeric(data_test$predict)
data_test$bench_pred <- as.numeric(data_test$bench_pred)

all_rmse <- sqrt(mean((data_test$rate -  data_test$predict)^2))
all_rmse_bench <- sqrt(mean((data_test$rate -  data_test$bench_pred)
   ^2))

all_mae <- mean(abs(data_test$rate -  data_test$predict))
```

```
all_mae_bench <- mean(abs(data_test$rate -  data_test$bench_pred))

## 90th percentile
obs_top10 <- data_test[data_test$rate >= quantile(data_test$rate ,
   0.90) ,]
obs_top10[order(obs_top10$rate ,decreasing=TRUE),c("FIPS_num","County
   ","rate")]#observed 90th percentile

pred_top10 <- data_test[data_test$predict >= quantile(
   data_test$predict , 0.90) ,]
pred_top10[order(pred_top10$predict ,decreasing=TRUE),c("FIPS_num","
   County","predict")]#predicted 90th percentile

rmse_top10 <- data_test[as.numeric(data_test$TestRMSE) >= quantile(as
   .numeric(data_test$TestRMSE), 0.90) ,]
rmse_top10[order(as.numeric(rmse_top10$TestRMSE),decreasing=TRUE),c("
   FIPS_num","County","TestRMSE","rate","predict")]#RMSE 90th
   percentile

pred_top10_bench <- data_test[data_test$bench_pred >= quantile(
   data_test$bench_pred , 0.90) ,]
pred_top10_bench[order(pred_top10_bench$bench_pred ,decreasing=TRUE),c
   ("FIPS_num","County","predict")]#predicted 90th percentile
```

5.2 Spatial Random Forest

To fit the spatial random forest model, I used the packages spatialRF and sf (Benito, 2025; Pebesma, 2018; Pebesma & Bivand, 2023; Wright & Ziegler, 2017).

This code chunk will read in the necessary packages, the data, and the shapefiles. It also defines the variables that store the information necessary to get the pooled MSE and MAE later.

```
source("FunctionsCntyDeath.R")
library(spatialRF)
library(sf)

choose_state <- c("WV") # west virginia
StateNameNum <- read.csv("StateNumToAbb.csv",colClasses = c(Number =
   "character"))
state_list <- StateNameNum %>% filter(Abbreviation %in% choose_state)
state_list <- c(state_list[,1])

test_yr = 2023
```

```
mse_year <- matrix(ncol=4,nrow=length(c(2015:2022)))
map_county <- sf::st_read("ShapeFiles/tl_2021_us_county.shp",
   stringsAsFactors=F,as_tibble=T)
map_county <- map_county %>% filter(substr(GEOID,1,2) %in% state_list
   )
coordinates <- st_coordinates(st_centroid(st_geometry(map_county),
   of_largest_polygon=TRUE) )[,1:2]
distance_matrix <- as.matrix(dist(coordinates))

coordinates <- as.data.frame(coordinates)
names(coordinates) <- c("x","y")

sum_sq_pred <- 0
sum_abs_pred <- 0
n_sum_sq <- 0

data <- getOverdoseData(state=choose_state,choose_year="all")
data <- data[data$Year >= 2015,]
#covar
dfCov <- read.csv("cntyCovar2015-2024.csv",colClasses = c(GEOID = "
   character"))
dfCov$Year <- as.numeric(dfCov$Year)
data <- left_join(data, dfCov, by=c("FIPS_num"="GEOID", "Year"))
```

This code chunk will loop through each year and fit the model rf_spatial() and save the rf_importance() plots. It will also calculate the pooled MSE and MAE.

```
for(i in 2015:2022)
{
  data_join <- left_join(map_county,data[data$Year == i,], by=c("
     GEOID"="FIPS_num"))

  rf_spatial_moran <- rf_spatial(
      data = data_join,
      dependent.variable.name = "rate",
      predictor.variable.names = c("Cognitive","SelfCare","Insured","
         Poverty","Education","maritalStatus_D_acs","medianIncome"),
      distance.matrix = distance_matrix,
      distance.thresholds = 0,
      method = "mem.moran.sequential",
      n.cores = 1,
      verbose=F#stop automatix outputs
  )

  rf_imp <- rf_importance(
    rf_spatial_moran,
    xy = coordinates
  )
```

```
  png(paste("rf_importance_",i,".png",sep=""), width = 300, height =
     200)
  print(plot_importance(rf_spatial_moran))

  dev.off()#must run after saving plots

  # https://blasbenito.github.io/spatialRF/articles/spatial_models.
     html: Spatial predictors are named spatial_predictor_X_Y, where
     X is the neighborhood distance at which the predictor is
     generated, and Y is the index of the predictor.
  # higher val, more important

  mse_year[(i-2014),1] <- i
  mse_year[(i-2014),2] <- mean((data_join$rate -
     rf_spatial_moran$predictions$values)^2)
  mse_year[(i-2014),3] <- sqrt(mse_year[(i-2014),2])
  mse_year[(i-2014),4] <- mean(abs(data_join$rate -
     rf_spatial_moran$predictions$values))#mean abs error

  sum_sq_pred <- sum((data_join$rate -
     rf_spatial_moran$predictions$values)^2)+sum_sq_pred
  sum_abs_pred <- sum(abs(data_join$rate -
     rf_spatial_moran$predictions$values))+sum_abs_pred
  n_sum_sq <- length(rf_spatial_moran$predictions$values)+n_sum_sq
}

all_rmse <- sqrt(sum_sq_pred/n_sum_sq)
all_rmse
all_mae <- sum_abs_pred/n_sum_sq
all_mae
mse_year <- as.data.frame(mse_year)
names(mse_year) <- c("Year","MSE","RMSE","MAE")
mse_year
```

5.3 Spatial Autoregression

To fit the SAR models, I used the sf, spdep, and spatialreg packages (R. Bivand, 2022; R. Bivand & Wong, 2018; R. S. Bivand et al., 2013; Pebesma, 2018; Pebesma & Bivand, 2023).

This code chunk, will load in the necessary packages, read the data in, and load in the shapefiles. It also defines the variables that store the information necessary to get the pooled MSE and MAE later.

```
library(sf)
library(spdep)
library(spatialreg) # spautolm

source("FunctionsCntyDeath.R")

#choose state(s) to run analysis on
StateNameNum <- read.csv("StateNumToAbb.csv",colClasses = c(Number =
   "character"))

choose_state <- "WV"
state_list <- StateNameNum %>% filter(Abbreviation %in% c(
   choose_state))
state_list <- c(state_list[,1])

map_county <- sf::st_read("ShapeFiles/tl_2021_us_county.shp",
   stringsAsFactors=F,as_tibble=T)
map_county <- map_county %>% filter(STATEFP %in% state_list)

data <- getOverdoseData(choose_state,"all")
data <- data[data$Year >= 2015,]
dfCov <- read.csv("cntyCovar2015-2024.csv",colClasses = c(GEOID = "
   character"))
dfCov$Year <- as.numeric(dfCov$Year)
data <- left_join(data, dfCov, by=c("FIPS_num"="GEOID", "Year"))

mse_year <- matrix(ncol=7,nrow=length(c(2015:2022)))

sum_sq_pred_1 <- 0
sum_abs_pred_1 <- 0
n_sum_sq_1 <- 0

sum_sq_pred <- 0
sum_abs_pred <- 0
n_sum_sq <- 0
```

This next code chunk will loop through each year in the training set and fit the SAR models with spautolm(). The neighbor matrix and weights matrix are made with poly.nb.narm(), poly2nb(), and nb2listw(). It will also output various information on the model, Moran's I statistics, and continue the calculations for MSE and MAE.

```
for(i in 2015:2022)
{
  map_county_join <- left_join(map_county,data[data$Year == i,],by=c
     ("GEOID"="FIPS_num"))

  poly.nb.narm <- poly2nb(map_county_join)
```

```
  W = nb2listw(poly.nb.narm, style="W",zero.policy=T)
  # coords <- st_centroid(st_geometry(map_county_join),
     of_largest_polygon=TRUE)
  # poly.nb <- poly2nb(map_county_join)
  # plot(st_geometry(map_county_join), border="grey")
  # plot(poly.nb, coords, add=TRUE)

  sar_1 = spautolm(rate~1, data=map_county_join,W, family="SAR") #for
      no covariates, intercept only

  sar = spautolm(rate~Cognitive+SelfCare+Insured+Poverty+Education+
     maritalStatus_D_acs+medianIncome,
                 data=map_county_join,W, family="SAR")

  cat("YEAR:",i,"---------------------\n")
  cat("INTERCEPT:",i,"\n")
  print(summary(sar_1))
  cat("COVAR:",i,"\n")
  print(summary(sar))
  cat("MORAN:",i,"\n")
  print(moran.test(map_county_join$rate,W))

  mse_year[(i-2014),1] <- i
  mse_year[(i-2014),2] <- mean((sar_1$fit$residuals)^2)
  mse_year[(i-2014),3] <- sqrt(mse_year[(i-2014),2])
  mse_year[(i-2014),4] <- mean(abs(sar_1$fit$residuals))#mean abs
     error
  mse_year[(i-2014),5] <- mean((sar$fit$residuals)^2)
  mse_year[(i-2014),6] <- sqrt(mse_year[(i-2014),5])
  mse_year[(i-2014),7] <- mean(abs(sar$fit$residuals))#mean abs error

  sum_sq_pred_1 <- sum((sar_1$fit$residuals)^2)+sum_sq_pred_1
  sum_abs_pred_1 <- sum(abs(sar_1$fit$residuals))+sum_abs_pred_1
  n_sum_sq_1 <- length(sar_1$fit$residuals)+n_sum_sq_1

  sum_sq_pred <- sum((sar$fit$residuals)^2)+sum_sq_pred
  sum_abs_pred <- sum(abs(sar$fit$residuals))+sum_abs_pred
  n_sum_sq <- length(sar$fit$residuals)+n_sum_sq

}

all_rmse_1 <- sqrt(sum_sq_pred_1/n_sum_sq_1)
all_rmse_1
all_mae_1 <- sum_abs_pred_1/n_sum_sq_1
all_mae_1

all_rmse <- sqrt(sum_sq_pred/n_sum_sq)
all_rmse
all_mae <- sum_abs_pred/n_sum_sq
all_mae
```

```
mse_year <- as.data.frame(mse_year)
names(mse_year) <- c("Year","mse_int","rmse_int","mae_int","mse","
   rmse","mae")
mse_year

res <- as.data.frame(matrix(c(sar$fit$residuals,names(
   sar$fit$residuals)),ncol=2))
names(res) <- c("res","row_num")
pred <- as.data.frame(matrix(c(sar$fit$fitted.values,names(
   sar$fit$fitted.values)),ncol=2))
names(pred) <- c("pred","row_num")

res1 <- as.data.frame(matrix(c(sar_1$fit$residuals,names(
   sar_1$fit$residuals)),ncol=2))
names(res1) <- c("res1","row_num")
pred1 <- as.data.frame(matrix(c(sar_1$fit$fitted.values,names(
   sar_1$fit$fitted.values)),ncol=2))
names(pred1) <- c("pred1","row_num")

map_county$row_num <- row.names(map_county)
map_county <- left_join(map_county,res,by="row_num")
map_county <- left_join(map_county,pred,by="row_num")
map_county$res <- as.numeric(map_county$res)
map_county$pred <- as.numeric(map_county$pred)

map_county <- left_join(map_county,res1,by="row_num")
map_county <- left_join(map_county,pred1,by="row_num")
map_county$res1 <- as.numeric(map_county$res1)
map_county$pred1 <- as.numeric(map_county$pred1)

mae <- mean(abs(sar$fit$residuals))
mae
mse <- mean((sar$fit$residuals)^2)
sqrt(mse)

mae1 <- mean(abs(sar_1$fit$residuals))
mae1
mse1 <- mean((sar_1$fit$residuals)^2)
sqrt(mse1)
```

5.4 Gaussian Process Boosting

To fit the GPBoost models I used the packages gpboost and sf (Pebesma, 2018; Pebesma & Bivand, 2023; Sigrist et al., 2025).

This code chunk will read in the necessary packages, the data, and the shapefiles. Some helpful websites for fitting GPBoost are linked in the comments of this code, and mentioned earlier in this paper (Sigrist, 2023a, 2023b).

```
library(gpboost)
library(sf) #st_centroid, st_coordinates

source("FunctionsCntyDeath.R")

# demonstration: https://medium.com/data-science/mixed-effects-
   machine-learning-for-longitudinal-panel-data-with-gpboost-part-iii
   -523bb38effc
# w/ coords demo: https://medium.com/data-science/mixed-effects-
   machine-learning-with-gpboost-for-grouped-and-areal-spatial-
   econometric-data-b26f8bddd385

#group_data is the ID for each group
#data is predictor variables, should include response, year, and any
   covariates
#label is name of response variable

set.seed(2025)
## choose state(s) to run analysis on
StateNameNum <- read.csv("StateNumToAbb.csv",colClasses = c(Number =
   "character"))

# choose_state <- c("ME","VT","NH","MA","RI","CT","NY","NJ","PA","DE
   ","MD", "DC") # northeast states, 4655
# choose_state <- c("GA","NC","SC","VA") # southeast coast, 6715
# choose_state <- c("AR","TN","WV","KY") # appalachian, 5075
# choose_state <- c("FL","AL","MS","LA") # gulf coast, 4358
# choose_state <- c("AZ","NM","OK","TX") #Southwest states, 5694
# choose_state <- c("WI","MI","OH","IL","IN") # great lakes, 6801
# choose_state <- c("ND","SD","NE","KS","MN","IA","MO") # midwest
   states, 8910
# choose_state <- c("WA","MT","ID","WY","CO","UT","NV","CA","OR") #
   western states, 5653

choose_state <- c("WV") # west Virginia

state_list <- StateNameNum %>% filter(Abbreviation %in% choose_state)
state_list <- c(state_list[,1])

# choose_state <- "all"
# state_list <- c(StateNameNum[c(1,3:11,13:51),1])#all

# get centroids of each state
map_county <- sf::st_read("ShapeFiles/tl_2021_us_county.shp",
   stringsAsFactors=F,as_tibble=T)
```

```
map_county <- map_county %>% filter(STATEFP %in% state_list)
centroids <- map_county %>% st_centroid()
centroids$long <- st_coordinates(centroids[,"geometry"])[,1]
centroids$lat <- st_coordinates(centroids[,"geometry"])[,2]

test_yr = 2023
data <- getOverdoseData(choose_state,"all")

### join covar data
dfCov <- read.csv("cntyCovar2015-2024.csv",colClasses = c(GEOID = "
   character"))
dfCov$Year <- as.numeric(dfCov$Year)
data <- left_join(data, dfCov, by=c("FIPS_num"="GEOID", "Year"))
data <- data[data$Year >= 2015,] # some covar are valid for 2015 to
   2024, some 2010 to 2024

### join map data
data <- inner_join(data,centroids,by=c("FIPS_num"="GEOID")) # want
   inner join

# data[is.na(data[,"long"]),] # check for issues in data. if a cnty
   does not have long or lat that is an issue.

data <- data[,-which(names(data)=="geometry")]   #remove geometry
data <- data[!is.na(data$rate),] # remove NA rate

data_train <- data.matrix(data[data$Year < test_yr,])
data_test <- data.matrix(data[data$Year == test_yr,])
```

There are three models included here, the GPBoost model with a spatial Gaussian process, a GPBoost model with a temporal Gaussian process by cluster (not discussed in write up), and a GPBoost model with a shared temporal Gaussian process.

This first chunk will define the model and get the data into the correct format. The code is set up to run the first model discussed above. If you want to run one of the other models, simply uncomment its code block and run it. Tip: to comment or uncomment many lines of R code in RStudio at once, use ctr+shift+c.

```
## FIT GPBOOST model
# Gaussian Process by group: Space, Random Effect: Year, Fixed Effect
   : -many- ##############3
rand_effect <- c("Year")
fixed_effect <- c("Cognitive", "SelfCare","Insured" ,"Poverty",
                  "Education" ,"maritalStatus_D_acs","medianIncome")

gp_model <- GPModel(group_data = data_train[,c("FIPS_num")], #grouped
```

```
    random effects, each group_data has its own intercept
                    group_rand_coef_data = data_train[,rand_effect],#
                       random  effects (slope) covars for group_data
                    ind_effect_group_rand_coef =c(seq(1,1,length.out
                       = length(rand_effect))),
                    gp_coords = data_train[,c("long","lat")],#
                       Gaussian process random effects
                    # cluster_ids = ,# independent spatial and
                       grouped random effects for each cluster but
                       shared fixed effects
                    likelihood = "gaussian",cov_function = "
                       exponential")#21.72 and 32.69
# #end

# Gaussian Process by cluster: Time, Random Effect: Year and
   Cognitive, Fixed Effect: Year, Cognitive, FIPS_num
# cognitive optional
# # each cluster_ids has a separate temporal Gaussian process (
   gp_coords)
# rand_effect <- c("Cognitive")
# fixed_effect <- c(rand_effect,"FIPS_num")
#
# gp_model <- GPModel(gp_coords = data_train[,c("Year")],
#                      group_rand_coef_data = data_train[,rand_effect
   ],#add covar
#                      ind_effect_group_rand_coef = c(seq(1,1,length.
   out = length(rand_effect))),
#                      likelihood = "gaussian",cov_function="
   exponential",
#                      cluster_ids = data_train[,"FIPS_num"])
# #end

# Gaussian Process (shared): Time, Fixed Effect:-many
   -##################3
# single temporal Gaussian process (gp_coords) shared across
   group_data
# rand_effect <- c("")
# fixed_effect <- c("Cognitive", "SelfCare","Insured" ,"Poverty",
#                    "Education" ,"maritalStatus_D_acs","medianIncome
   ")
#
# gp_model <- GPModel(group_data=data_train[,"FIPS_num"],
#                      # group_rand_coef_data = data_train[,
   rand_effect],#add covar
#                      # ind_effect_group_rand_coef = c(seq(1,1,length
   .out = length(rand_effect))),
#                      gp_coords = data_train[,c("Year")],
#                      likelihood = "gaussian",cov_function="
   exponential")
#end
```

```
### end model definitions
#################################################################

## make data set
data_bst <- gpb.Dataset(data = data_train[,c(fixed_effect)], #fixed
   effects
                     label = data_train[,"rate"])#response data
```

This second chunk will train the GPBoost model and output the importance plots. Again, this code is set up to run the fist model discussed. This chunk also includes the code to optimize the tuning parameters. Which I already did, and now have those parameters stored in the variables params and nrounds.

```
####### TRAIN MODEL #####

# Gaussian Process by cluster (I did not find optimal tuning
   parameters for this model)
# params <- list(learning_rate = 0.01, max_depth = 2, num_leaves =
   2^10,
#                min_data_in_leaf = 10, lambda_l2 = 10)#old
# nrounds <- 5 #37  # number of boosting iterations

# Find optimal tuning parameters
# opt_params <- gpb.grid.search.tune.parameters(param_grid =
   param_grid, params = other_params,
#                                                num_try_random = NULL
   , folds = folds,
#                                                data = data_bst,
   gp_model = gp_model,
#                                                nrounds = 1000,
   early_stopping_rounds = 10,
#                                                verbose_eval = 1,
   metric = "mse")
# opt_params

# Gaussian Process by group
params <- list(learning_rate = 1, max_depth = 10, num_leaves = 2^10,
              min_data_in_leaf = 10, lambda_l2 = 0)#best for space
nrounds <- 880#best for space, too much for the 8 regions
# nrounds <- 40 # for space all usa regions
#end

# Gaussian Process (shared): Time
# params <- list(learning_rate = 1, max_depth = 10, num_leaves =
   2^10,
#                min_data_in_leaf = 10, lambda_l2 = 10)#best for time
```

```
# nrounds <- 90#is best for time# number of boosting iterations
#end

gpbst <- gpb.train(data = data_bst, gp_model = gp_model, nrounds =
   nrounds,
                   params = params, verbose = 0)
#gpboost() below  is the same as gpb.train() above
# gpbst <- gpboost(data = data_bst, gp_model = gp_model,nrounds =
   nrounds, params = params, verbose = 1) #same as gpb.train() above

# look at model outputs
summary(gp_model)# Estimated covariance parameters
var(data_train[,"rate"]) #variance of response
# predict_training_data_random_effects(gp_model)

View(gpb.importance(gpbst))
# higher "Gain" means the covar is more important for reducing model
   error
feature_importances <- gpb.importance(gpbst, percentage = TRUE)
gpb.plot.importance(feature_importances,  measure = "Gain",
                    main = "Split-based variable importances")
```

This third chunk will get the predictions from the model for the training set. It is set up to run the first model discussed here. This code will also calculate the MSE and MAE.

```
####### TRAINING PREDICTION #####
# Gaussian Process by group
train_pred <- predict(gpbst, data = data_train[,c(fixed_effect)],
                gp_coords_pred = data_train[, c("long", "lat")],
                group_data_pred = data_train[,c("FIPS_num")],
                group_rand_coef_data_pred = data_train[,rand_effect],
                # cluster_ids_pred = ,
                predict_var = TRUE)
# # end

# Gaussian Process by cluster
# train_pred <- predict(gpbst, data = data_train[,c(fixed_effect)],
#                  gp_coords_pred = data_train[, "Year"],
#                  cluster_ids_pred = data_train[,"FIPS_num"],
#                  group_rand_coef_data_pred = data_train[,rand_effect
   ],
#                  predict_var = TRUE)
# #end

# Gaussian Process (shared): Time
# train_pred <- predict(gpbst, data = data_train[,c(fixed_effect)],
#                  group_data_pred = data_train[,c("FIPS_num")],
#                  gp_coords_pred = data_train[, "Year"],
#                  # group_rand_coef_data_pred = data_train[,
```

```
   rand_effect],
#                  predict_var = TRUE)
#end
##############################################################

train_mse <- mean((data_train[,"rate"] - train_pred$response_mean)^2)
sqrt(train_mse)
train_mae <- mean(abs(data_train[,"rate"] - train_pred$response_mean)
   )
train_mae
```

This last chunk will run the model on the test set and calculate the MSE and MAE.

```
##############################################################
####### TEST PREDICTION #####
# Gaussian Process by group
pred <- predict(gpbst, data = data_test[,c(fixed_effect)],
                gp_coords_pred = data_test[, c("long", "lat")],
                group_data_pred = data_test[,"FIPS_num"],
                group_rand_coef_data_pred = data_test[,rand_effect],
                predict_var = TRUE)

# Gaussian Process by cluster
# pred <- predict(gpbst, data = data_test[,c(fixed_effect)],
#                  gp_coords_pred = data_test[, "Year"],
#                  cluster_ids_pred = data_test[,"FIPS_num"],
#                  group_rand_coef_data_pred = data_test[,rand_effect
   ],
#                  predict_var = TRUE)
# # end

# Gaussian Process (shared): Time
# pred <- predict(gpbst, data = data_test[,c(fixed_effect)],
#                  group_data_pred = data_test[,c("FIPS_num")],
#                  gp_coords_pred = data_test[, "Year"],
#                  # group_rand_coef_data_pred = data_test[,
   rand_effect],
#                  predict_var = TRUE)
# end
##############################################################

y_pred <- pred$response_mean #test set prediction
test_mse <- mean((data_test[,"rate"] - y_pred)^2)
sqrt(test_mse)
test_mae <- mean(abs(data_test[,"rate"] - y_pred))
test_mae

data$pred[data$Year == test_yr] <- pred$response_mean #test set
   prediction
data$pred[data$Year < test_yr] <- train_pred$response_mean
```

```
data$var[data$Year == test_yr] <- pred$response_var
data$res <- data$rate - data$pred

## 90th percentile
data_2023 <- data[data$Year == test_yr,]
mean(data_2023$pred)#predicted mean
pred_top10 <- data_2023[data_2023$pred >= quantile(data_2023$pred,
   0.90),]
pred_top10[order(pred_top10$pred,decreasing=TRUE),c("FIPS_num","
   County","pred")]#predicted 90th percentile
```

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